

# LARGE ERDŐS–STRAUS WITNESSES II: THE MULTIPLIER MODULUS IS INFINITELY OFTEN $\exp((\log p)^{1/5-o(1)})$

ANONYMOUS

**ABSTRACT.** For a prime  $p$  let  $W(p)$  be the least modulus  $M \equiv 3 \pmod{4}$  of a “multiplier” congruence that yields a solution of  $4/p = 1/x + 1/y + 1/z$ . In earlier work we showed that  $W(p)$  exceeds every fixed power of  $\log p$  for infinitely many primes  $p$  in Mordell’s hard classes. Here we prove  $W(p) \geq \exp(c(\log p)^{1/5}(\log \log p)^{-1/5})$  for infinitely many hard primes, assuming Gallagher’s prime number theorem for the family of all characters and four results of Elsholtz and Tao on divisor sums of polynomials; more precisely, the least hard prime with  $W(p) > T$  is at most  $\exp(O((\log T)^5 \log \log T))$ . Without the Elsholtz–Tao input we get  $\log W(p) \geq (1/\log 2 - o(1)) \log \log p \cdot \log \log \log p$  infinitely often. The class of one modulo a *graded* set of prime powers leaves a residual congruence system; the exponent at each prime is raised until the mass of the system at that prime falls below a threshold proportional to the logarithm of the prime power it would consume. The local lemma then gives its avoiding set positive Haar density, and the cost of the quarantine is controlled by a moment of the number of prime-power divisors of the moduli, which we bound with the Elsholtz–Tao divisor sums in progressions. To transfer the density to primes we use the one-sided  $\ell^2$  sandwich of Bazzi and Razborov and a linear transfer theorem based on Gallagher’s theorem, as in the previous version. The new ingredient is a sharp energy bound for avoidance indicators: for any family of cylinder events on a finite product probability space and any weights  $\lambda_v \geq 1$  with  $\prod_{v \in \text{supp } E} \lambda_v \leq 2$  for every event  $E$ , the Efron–Stein energies satisfy  $\sum_U \prod_{v \in U} \lambda_v \|F^{=U}\|_2^2 \leq 1$ . It replaces the binary encoding, Håstad’s switching lemma and the Linial–Mansour–Nisan argument of the previous version, and, applied to the base- $\ell$  digits of the residues, it measures the cost of the approximation in the size of the moduli instead of the number of primes.

## 1. INTRODUCTION

The Erdős–Straus conjecture asserts that  $4/n = 1/x + 1/y + 1/z$  is solvable in positive integers for every  $n \geq 2$ . It suffices to treat primes. It has been verified for  $n \leq 10^{18}$  ([25]; [31] for  $10^{17}$ ), and Mordell’s identities settle every prime outside the six classes  $p \equiv 1^2, 11^2, 13^2, 17^2, 19^2, 23^2 \pmod{840}$  [24] (cf. [27]). We call the primes in these classes *hard*. For the exceptional set see Vaughan [34] and [28]; for the average number of solutions see [10].

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For  $M \equiv 3 \pmod{4}$  put  $A_M = (M+1)/4$ . If  $uvw = A_M$  and  $n \equiv -uv^{-1} \pmod{M}$ , then  $s = (nv+u)/M$  is a positive integer and

$$\frac{4}{n} = \frac{1}{suw} + \frac{1}{nsuv} + \frac{1}{nuvw} \quad (1)$$

(direct verification; this is the classical multiplier family, cf. [34, 10]). Define

$$W(n) = \min\{M \equiv 3 \pmod{4} : n \equiv -uv^{-1} \pmod{M} \text{ for some } uvw = A_M\}, \quad (2)$$

with  $W(n) = +\infty$  if no such  $M$  exists. Thus  $W(p) \leq T$  provides an explicit representation with modulus at most  $T$ . Every prime  $p \equiv 1 \pmod{\text{lcm}\{M \leq T\}}$  has  $W(p) > T$  (Lemma 2.2), so Linnik's theorem gives  $W(p) \gg \log p$  infinitely often. In [2] we proved that for every fixed  $k$ ,  $W(p) \geq (\log p)^k e^{-C_k \log \log p / \log \log \log p}$  for infinitely many hard primes. Here we prove a sub-exponential rate.

**Theorem 1.1** (Proved modulo Theorems 4.2 and 3.1). *There is an absolute constant  $c > 0$  such that for infinitely many hard primes  $p$*

$$W(p) \geq \exp\left(c \frac{(\log p)^{1/5}}{(\log \log p)^{1/5}}\right).$$

*More precisely, for every sufficiently large  $T$  there is a prime  $p \equiv 1 \pmod{840}$  with  $p > T$ ,  $W(p) > T$  and  $\log p \leq C(\log T)^5 \log \log T$ .*

**Theorem 1.2** (Proved modulo Theorem 4.2). *For infinitely many hard primes  $p$ ,*

$$\log W(p) \geq \left(\frac{1}{\log 2} - o(1)\right) \log \log p \cdot \log \log \log p.$$

Theorem 4.2 is Gallagher's prime number theorem for the family of all primitive characters of conductor at most  $Q_G$  [12], quoted from [22, Thm. 28.19] together with the standard facts on exceptional zeros (Theorem 4.1). No hypothesis on Siegel zeros is needed. Theorem 3.1 collects four results of Elsholtz and Tao [10]: their Proposition 1.4, Theorem 7.1, Corollary 7.4 and the estimate (7.10). They are used only in §3, to bound the total mass of the residual system by  $O((\log T)^4)$  and a moment of the number of prime-power divisors of its moduli by  $O((\log T)^4 \log \log T)$ . Without them the same argument gives Theorem 1.2. All constants are effective, except that we have not checked whether the implied constants in [10] are. We make no numerical claim. We also obtain a Haar-side bound (Corollary 8.2): the set of integers  $n$  with  $W(n) > T$  has density at least  $\exp(-O((\log T)^5 \log \log T))$ , modulo Theorem 3.1.

*Versions.* The first version of this note had exponent  $1/14$ ; it used the Thorner–Zaman theorem [32] through the transfer theorem of [2]. The second had exponent  $1/7$ : it replaced Thorner–Zaman by the linear transfer theorem (Theorem 5.1 here, in a simpler form), and bounded the approximation error by Håstad's switching lemma. The present version keeps the transfer theorem and the sandwich, and changes three things: the switching lemma is replaced

by an energy bound (§6), the quarantine is graded (§2), and the cost of the quarantine is bounded through a moment estimate (§3). Håstad's switching lemma is no longer used.

**The method.** Fix  $T$ . We impose  $n \equiv 1$  modulo  $Q = 8 \prod_{\ell} \ell^{a_{\ell}}$  for suitable exponents  $a_{\ell} \geq 0$ . Then  $W(n) > T$  whenever  $n$  avoids a finite family of congruence events  $n \equiv -4D \pmod{r}$ , one for each surviving pair  $(M, D)$ , where  $r$  is the part of  $M$  not absorbed by  $Q$ . Unlike in [2] and the previous version, a prime  $\ell$  is not quarantined all at once to  $n \equiv 1 \pmod{\ell^e}$  with  $\ell^e \leq T < \ell^{e+1}$ . We raise  $a_{\ell}$  one step at a time, as long as the Haar mass  $w_{\ell}$  of the events that still involve  $\ell$  exceeds  $\theta_{\ell} = c(a_{\ell} + 1) \log \ell / \log T$  (Lemma 2.5). Since every event comes from a modulus  $M \leq T$ , the total mass seen by one event is then at most  $c$ , which is what the local lemma needs. The cost  $\log Q$  is at most  $(\log T)/c$  times the moment  $\sum_{(M,D)} (g/M) h(M)$ , where  $g = \gcd(M, 4D + 1)$  and  $h(M) = \sum_{\ell^i | M} 1/i$ . We bound this moment by  $O((\log T)^4 \log \log T)$  (Theorem 3.7), so  $\log Q \ll (\log T)^5 \log \log T$ .

To find a *prime* in the avoiding set we need a function  $B = \sum_i c_i \mathbf{1}[n \equiv b_i \pmod{d_i}]$  with  $B \leq \mathbf{1}[\text{avoid}]$  pointwise, mean  $\gg \delta$  (the Haar density of the avoiding set), Haar  $\ell^1$  norm  $\mathbb{E}|B|$  comparable to its mean, and small moduli  $d_i$ . The linear transfer theorem (Theorem 5.1) then gives a prime  $p$  with  $\log p \ll \log Z$ , where  $Z$  is  $Q$  times the largest  $d_i$ . It is Linnik's theorem with weights: the function  $B(n) \mathbf{1}[n \equiv 1 \pmod{Q}]$  is expanded in characters, every coefficient is bounded by the Haar  $\ell^1$  norm of  $B$ , whatever the number of cells, and Gallagher's theorem bounds the contribution of all characters of conductor at most a power of  $Z$  at once.

As in the previous version,  $B$  is the one-sided  $\ell^2$  sandwich of Bazzi [3], in the form of Razborov [29] (Lemma 7.4). For events  $E_1, \dots, E_m$  with indicators  $A_i$  and arbitrary functions  $u_j$ ,

$$B = 1 - \sum_i A_i \left( 1 - \sum_{j < i} A_j u_j \right)^2 \leq \mathbf{1}[\text{no } E_i \text{ occurs}],$$

and  $\mathbb{E}[F - B] \leq m^2 \sum_j \mathbb{P}(E_j) \mathbb{E}(F^{(j)} - u_j)^2$ , where  $F^{(j)}$  is the indicator that no earlier event occurs, conditioned on  $E_j$ . We need the  $u_j$  to be combinations of classes with small moduli. The natural choice is a truncation of the Efron–Stein decomposition of  $F^{(j)}$ , and the error is then an Efron–Stein tail. The previous version bounded this tail by encoding residues in binary and applying Håstad's switching lemma [15, 20]. Here we prove (Theorem 6.5): if  $F$  is the avoidance indicator of any family of cylinder events on a finite product probability space, and  $\lambda_v \geq 1$  are weights with  $\prod_{v \in \text{supp } E} \lambda_v \leq 2$  for every event  $E$ , then

$$\sum_U \left( \prod_{v \in U} \lambda_v \right) \|F^{=U}\|_2^2 \leq 1.$$

With  $\lambda_v = 2^{1/k}$  this gives the tail  $2^{-(t+1)/k}$  beyond level  $t$  for events on at most  $k$  coordinates (Corollary 6.6), with no dependence on the alphabet

sizes. Applied to the base- $\ell$  digits of the residues, with weights adapted to the moduli (Theorem 6.9), it shows that truncating  $F^{(j)}$  to moduli at most  $e^\tau$  costs  $2^{-\tau/(2\log T)}$ . So moduli of size  $\log d_i \ll \log T \cdot (S + \log T)$  suffice, where  $S \ll (\log T)^4$  is the total mass. Altogether  $\log Z \ll (\log T)^5 \log \log T$ .

**Relation to the literature.** *The energy bound.* The nearest prior art known to us is Lecomte and Tan [19], who bound the Fourier coefficients of a DNF on  $\{\pm 1\}^n$  by the probability that a set  $S$  is covered by the terms satisfied at a random point; this is the device of our Lemma 6.2. Their cover counts are unsigned, and their degree concentration still comes from Håstad’s switching lemma, with an unspecified constant. In Lemma 6.2 the cover counts are signed, so that they vanish when a satisfied event misses the set; the polarization identity (Lemma 6.3) and the deletion–contraction bound (Lemma 6.4) then give the constant 1, on any finite product probability space. In Boolean language (Remark 6.8): every width- $k$  DNF  $g : \{\pm 1\}^n \rightarrow \{\pm 1\}$  has  $\sum_{|S|>t} \hat{g}(S)^2 \leq 4 \cdot 2^{-(t+1)/k}$ , under any product measure, while the switching-lemma route gives  $2 \cdot 2^{-t/(20k)}$  [26, §4.4]. The base  $2^{-1/k}$  is sharp over general product spaces (Remark 6.7). Theorem 6.5 and its corollaries are new to us, but our literature search was partial (we checked [19], [26] and some lecture notes; we did not search the literature on switching-lemma variants, Fourier growth or hypercontractivity with  $\rho > 1$  systematically), and we make no priority claim.

*The rest.* The sandwich (Lemma 7.4) is Razborov’s identity. To our knowledge tools from the theory of bounded independence (“polylogarithmic independence fools DNF and  $\text{AC}^0$ ” [3, 29, 6]) have not been used before to construct sieve minorants for primes. Gallagher’s theorem is classically the engine of Linnik’s theorem on the least prime in a single progression. Here it serves as a transfer device for an arbitrary signed combination of progressions; the cost is governed by the Haar  $\ell^1$  norm of the minorant and its largest modulus, not by the number of cells. The nearest classical analogues are bounds for the least prime in a union of residue classes, or in a Chebotarev class of an abelian extension (Lagarias–Montgomery–Odlyzko [23]; Thorner–Zaman [33]); these concern nonnegative indicators. We have not searched systematically for an earlier use of Gallagher’s theorem in this role. Earlier uses of the switching lemma in number theory known to us run in the opposite direction: they show that the Möbius function is uncorrelated with bounded-depth circuits [13, 5]. The Haar side, a local-lemma lower bound for the density of the uncovered set of a congruence system, has relatives in the work on covering systems: Filaseta, Ford, Konyagin, Pomerance and Yu [11], who to our knowledge already use the local lemma for this purpose, Hough [16] and Balister, Bollobás, Morris, Sahasrabudhe and Tiba [4]. We do not use those results. The moment bound of §3 uses the parametrisation and the divisor-sum estimates of Elsholtz and Tao [10]; the splitting of the prime powers at the scale of the parameter  $c$  (Lemma 3.5) is ours.

**Status conventions.** Every theorem header carries a label. *Proved* means proved in full here. *Proved modulo X* means proved here assuming the cited statement X, whose statement we checked in the source but whose proof we did not check. *Cited* marks external statements; *Classical* marks standard results for which we include a proof or a precise textbook reference. The cited inputs of the main theorems are Gallagher’s theorem (Theorem 4.2, used together with the classical Landau–Page facts of Theorem 4.1) and the Elsholtz–Tao estimates (Theorem 3.1). In the labels, “modulo Theorem 4.2” includes Theorem 4.1. Classical results whose proof we include, or which are standard textbook facts (the local lemma, the Efron–Stein decomposition, Mertens’ theorem, the divisor bound, the Rosser–Schoenfeld bound for  $\vartheta(x)$ ), are not listed in the labels. *Assessment* marks heuristic comparisons, which are not claims. Nothing here bears on the truth of the Erdős–Straus conjecture: a prime with large  $W(p)$  may well have other representations.

**Notation.**  $T$  is large and  $\mathcal{L} = \log T$ ;  $\log_2$  always denotes the binary logarithm (iterated logarithms are written out as  $\log \log$ ).  $\tau$  is the divisor function and  $\tau^*(x) = \max_{n \leq x} \tau(n)$ . The letter  $q$  denotes a prime power  $\ell^i$  with  $i \geq 1$ , and  $\sum_q$  is a sum over prime powers. For  $\lambda = (\lambda_v)$  and a finite set  $U$  we write  $\lambda^U = \prod_{v \in U} \lambda_v$ .

## 2. ATOMS AND THE GRADED QUARANTINE

For  $M \equiv 3 \pmod{4}$  put  $\mathcal{R}(M) = \{-4D \bmod M : D \mid A_M^2\}$ . Note that  $\gcd(A_M, M) = 1$  (as  $4A_M = M + 1$ ), so every element of  $\mathcal{R}(M)$  is a unit class.

**Lemma 2.1** (Proved; cf. [2]).  $\{-uv^{-1} \bmod M : uvw = A_M\} = \mathcal{R}(M)$ . Consequently, for every integer  $n$ ,  $W(n) > T$  if and only if  $n \bmod M \notin \mathcal{R}(M)$  for every  $M \leq T$  with  $M \equiv 3 \pmod{4}$ .

*Proof.* Since  $4A_M \equiv 1 \pmod{M}$ , from  $uvw = A_M$  we get  $v^{-1} \equiv 4uw$ , hence  $-uv^{-1} \equiv -4u^2w$  and  $D = u^2w$  divides  $A_M^2$ . Conversely let  $D \mid A_M^2$ . For each prime  $q$  with  $q^a \parallel A_M$  and  $q^d \parallel D$  ( $d \leq 2a$ ) put  $u_q = \max(0, d - a)$ ,  $w_q = d - 2u_q$ ,  $v_q = a - u_q - w_q$ ; these are non-negative, so  $u = \prod q^{u_q}$ ,  $v = \prod q^{v_q}$ ,  $w = \prod q^{w_q}$  satisfy  $uvw = A_M$  and  $u^2w = D$ .  $\square$

**Lemma 2.2** (Class of one; proved).  $1 \notin \mathcal{R}(M)$  for every  $M \equiv 3 \pmod{4}$ .

*Proof.* If  $-4D \equiv 1$  then  $D \equiv -A_M \pmod{M}$  (multiply by  $A_M$ ). Since  $4A_M^2 \equiv A_M$ , the map  $D \mapsto A_M^2/D$  preserves this congruence, so we may assume  $D \leq A_M$ ; then  $0 < D + A_M \leq 2A_M < M$ , while  $M \mid D + A_M$ .  $\square$

An *atom* is a pair  $(M, D)$  with  $M \leq T$ ,  $M \equiv 3 \pmod{4}$  and  $D \mid A_M^2$ . There are at most  $T^2$  atoms, since  $\tau(A_M^2) \leq 2A_M \leq T$ . For an atom put  $g = g(M, D) = \gcd(M, 4D + 1)$ .

**Definition 2.3** (Graded quarantine and residual system). For an odd prime  $\ell \leq T$  let  $f_\ell = \lfloor \mathcal{L} / \log \ell \rfloor \geq 1$  be the largest exponent with  $\ell^{f_\ell} \leq T$ . A *graded*

*quarantine* is a choice of integers  $0 \leq a_\ell \leq f_\ell$  for the odd primes  $\ell \leq T$ ; put  $Q = 8 \prod_\ell \ell^{a_\ell}$ . The *coordinate primes* are the odd  $\ell \leq T$  with  $a_\ell < f_\ell$ ; write  $\mathcal{P}$  for their set. For  $\ell \in \mathcal{P}$  let

$$\Omega_\ell = \{x \in (\mathbb{Z}/\ell^{f_\ell})^\times : x \equiv 1 \pmod{\ell^{a_\ell}}\}$$

(all units if  $a_\ell = 0$ ), with the uniform measure, and let  $\mathbb{P}, \mathbb{E}$  be the product measure on  $\Omega = \prod_{\ell \in \mathcal{P}} \Omega_\ell$ , with coordinates  $X_\ell$ . An atom *survives*  $Q$  if  $\gcd(M, Q) \mid 4D + 1$  and  $M \nmid Q$ . Its *support* is  $\text{supp}(M, D) = \{\ell : v_\ell(M) > a_\ell\}$ , which is a nonempty subset of  $\mathcal{P}$ , its *modulus* is  $r = \prod_{\ell \in \text{supp}} \ell^{v_\ell(M)}$ , and its *event* is

$$E_{M,D} = \{X_\ell \equiv -4D \pmod{\ell^{v_\ell(M)}} \text{ for all } \ell \in \text{supp}(M, D)\}.$$

The *fibre mass* at  $\ell \in \mathcal{P}$  and the total mass are

$$w_\ell(Q) = \sum_{\substack{(M,D) \text{ surviving} \\ \ell \in \text{supp}(M,D)}} \mathbb{P}(E_{M,D}), \quad S(Q) = \sum_{\substack{(M,D) \\ \text{surviving}}} \mathbb{P}(E_{M,D}),$$

where surviving atoms are counted individually (two atoms may give the same event). For an integer  $n$  we put  $X_\ell(n) = n \bmod \ell^{f_\ell}$ , and  $F(n) = 1$  if  $n \not\equiv -4D \pmod{r}$  for every surviving atom,  $F(n) = 0$  otherwise; on  $\Omega$ ,  $F$  is the indicator that no event  $E_{M,D}$  occurs.

Every event is nonempty:  $-4D$  is a unit modulo  $\ell$ , and  $-4D \equiv 1 \pmod{\ell^{a_\ell}}$  by survival. The modulus satisfies  $r \leq M \leq T$ . The quarantine of [2] and of the previous version is the special case  $a_\ell \in \{0, f_\ell\}$ .

**Lemma 2.4** (Reduction and weights; proved). *In Definition 2.3:*

- (i) *If  $n \equiv 1 \pmod{Q}$  and  $F(n) = 1$ , then  $W(n) > T$ .*
- (ii) *For every surviving atom,*

$$\mathbb{P}(E_{M,D}) = \frac{\gcd(M, Q)}{M} \prod_{\ell \in \text{supp}(M,D), a_\ell=0} \frac{\ell}{\ell-1} \leq \frac{g}{M} \prod_{\ell \mid M, a_\ell=0} \frac{\ell}{\ell-1}.$$

- (iii) *If  $\mathcal{L} \geq 2$  and  $a_\ell \geq 1$  for every odd prime  $\ell \leq \mathcal{L}$ , then  $\mathbb{P}(E_{M,D}) \leq C_1 g/M$  for every surviving atom, where  $C_1 = e^2$ .*

*Proof.* (i) Let  $M \leq T$ ,  $M \equiv 3 \pmod{4}$ , and suppose  $n \equiv -4D \pmod{M}$  with  $D \mid A_M^2$ . Reducing modulo  $\gcd(M, Q)$  gives  $-4D \equiv 1$ , i.e.  $\gcd(M, Q) \mid 4D + 1$ . If  $M \mid Q$ , then  $n \equiv 1 \pmod{M}$ , which contradicts Lemma 2.2. Otherwise the atom survives, and  $n \equiv -4D \pmod{r}$  as  $r \mid M$ , so  $F(n) = 0$ . Conclude with Lemma 2.1. (ii) Let  $\ell \in \text{supp}$  and  $v = v_\ell(M)$ . If  $a_\ell = 0$ , the class of  $-4D$  modulo  $\ell^v$  has probability  $1/\varphi(\ell^v) = \ell^{-v} \cdot \ell/(\ell-1)$  in  $\Omega_\ell$ . If  $a_\ell \geq 1$ , then  $|\Omega_\ell| = \ell^{f_\ell - a_\ell}$ , and the class (which lies in  $\Omega_\ell$ ) has  $\ell^{f_\ell - v}$  elements, so probability  $\ell^{a_\ell - v}$ . As  $M$  is odd,  $\gcd(M, Q) = \prod_{\ell \mid M} \ell^{\min(v_\ell(M), a_\ell)}$ , and  $\min(v_\ell(M), a_\ell) = v_\ell(M)$  for  $\ell \notin \text{supp}$ . This gives the identity. Survival gives  $\gcd(M, Q) \mid g$ . (iii) Every prime  $\ell$  with  $\ell \mid M$  and  $a_\ell = 0$  satisfies  $\ell > \mathcal{L}$ , so  $\ell/(\ell-1) \leq 1 + 1/(\mathcal{L}-1)$ . As  $M$  is odd,  $\omega(M) \leq \log M / \log 3 \leq \mathcal{L}$ . The product in (ii) is therefore at most  $\exp(\mathcal{L}/(\mathcal{L}-1)) \leq e^2$ .  $\square$

Write  $h(M) = \sum_{\ell^i | M, i \geq 1} 1/i = \sum_{\ell | M} H_{v_\ell(M)}$ , where  $H_v = \sum_{i \leq v} 1/i$ , and put

$$S_0 = \sum_{(M,D)} \frac{g}{M}, \quad \Omega_0 = \sum_{(M,D)} \frac{g}{M} h(M),$$

both sums over all atoms.

**Lemma 2.5** (Graded iterated quarantine; proved). *Let  $\mathcal{L} \geq 2$  and  $0 < c \leq 1/8$ . Start from  $a_\ell = 1$  for the odd primes  $\ell \leq \max(\mathcal{L}, 7)$  and  $a_\ell = 0$  for all other odd  $\ell$ . While some  $\ell \in \mathcal{P}$  has*

$$w_\ell(Q) > \theta_\ell := c(a_\ell + 1) \frac{\log \ell}{\mathcal{L}},$$

*raise  $a_\ell$  by one. The procedure stops, and the final quarantine satisfies:*

- (a)  $\sum_{\ell \in \text{supp}(M,D)} w_\ell(Q) \leq c$  for every surviving atom, and  $w_\ell(Q) \leq c$  for every  $\ell \in \mathcal{P}$ ;
- (b)  $\mathbb{P}(E_{M,D}) \leq C_1 g/M$  for every surviving atom, and  $S(Q) \leq C_1 S_0$ ;
- (c)  $840 \mid Q$  and  $\log Q \leq 1.02 \mathcal{L} + 7 + (C_1 \mathcal{L}/c) \Omega_0$ .

*Proof.* Each step raises some  $a_\ell < f_\ell$ , so the procedure stops. Exponents only grow, so Lemma 2.4(iii) applies at every stage, which gives (b). (a): at the end every  $\ell \in \mathcal{P}$  has  $w_\ell \leq \theta_\ell$ . If  $\ell \in \text{supp}(M, D)$ , then  $\ell \in \mathcal{P}$  and  $a_\ell + 1 \leq v_\ell(M)$ , so

$$\sum_{\ell \in \text{supp}(M,D)} w_\ell \leq \frac{c}{\mathcal{L}} \sum_{\ell | M} v_\ell(M) \log \ell = \frac{c \log M}{\mathcal{L}} \leq c.$$

Also  $w_\ell \leq \theta_\ell \leq c f_\ell \log \ell / \mathcal{L} \leq c$ , as  $a_\ell < f_\ell$  and  $\ell^{f_\ell} \leq T$ . (c)  $840 = 8 \cdot 3 \cdot 5 \cdot 7$  divides the starting  $Q$ , whose logarithm is at most  $\log(8 \cdot 105) + \vartheta(\mathcal{L}) \leq 1.02 \mathcal{L} + 7$  by  $\vartheta(x) < 1.01624x$  [30]. A step at  $\ell$  from  $a$  to  $a+1$ , taken at a stage with quarantine  $Q'$ , has  $\log \ell < \mathcal{L} w_\ell(Q') / (c(a+1))$ , and by (b) at that stage,  $w_\ell(Q') \leq C_1 \sum_{(M,D): v_\ell(M) \geq a+1} g/M$ . Each pair  $(\ell, a)$  is stepped at most once. Summing over all steps,

$$\log Q - \log Q_{\text{start}} \leq \frac{C_1 \mathcal{L}}{c} \sum_{(M,D)} \frac{g}{M} \sum_{\ell | M} \sum_{a=0}^{v_\ell(M)-1} \frac{1}{a+1} = \frac{C_1 \mathcal{L}}{c} \Omega_0. \quad \square$$

The local lemma needs only that the total mass seen by each event is small (Corollary 7.3), not that each  $w_\ell$  is at most a fixed fraction of  $1/k$ , where  $k$  bounds the number of primes in a support. Charging the threshold to the prime power  $\ell^{a_\ell+1}$  that a coordinate consumes lets the budget  $\log M \leq \mathcal{L}$  do the work of the width  $k$ . In particular no block  $\{\ell \leq z\}$  of fully quarantined small primes is needed. The ratio of the cost of a step to its threshold is  $\mathcal{L}/(c(a+1))$ , uniformly in  $\ell$ .

## 3. THE MASS AND THE CHARGE MOMENT

In this section we bound  $S_0$  and  $\Omega_0$ . We use four results of Elsholtz and Tao. For a polynomial  $P$  and  $m \geq 1$  let  $\rho_P(m)$  be the number of roots of  $P$  in  $\mathbb{Z}/m\mathbb{Z}$ , and for integers  $\kappa, a \geq 1$  let  $\rho_{\kappa a}(m)$  be the number of  $x \bmod m$  with  $\kappa ax^2 + 1 \equiv 0 \pmod{m}$ .

**Theorem 3.1** (Elsholtz–Tao [10]; cited). (a) [10, Prop. 1.4]: for  $A, B > 1$  and each integer  $\kappa \geq 1$  with  $\kappa \ll (AB)^{O(1)}$ ,  $\sum_{a \leq A} \sum_{b \leq B} \tau(\kappa ab^2 + 1) \ll AB \log(A+B) \log(1+\kappa)$ .  
 (b) ([10, Thm. 7.1]) Let  $N > 1$ , and let  $P$  be a polynomial of degree  $D_P$  whose coefficients are non-negative integers at most  $N^l$ , with  $\rho_P(p^j) \leq C$  for all primes  $p$  and all  $j \geq 1$ . Then  $\sum_{n \leq N} \tau(P(n)) \ll_{D_P, l, C} N \sum_{m \leq N} \rho_P(m)/m$ .  
 (c) ([10, Cor. 7.4]) If  $a, b, N$  are natural numbers with  $a, b \ll N^{O(1)}$ , then  $\sum_{n \leq N} \tau(an+b) \ll \tau(\gcd(a, b)) N \log N$ .  
 (d) ([10, (7.10)]) For  $2 \leq A \leq B$  and  $\kappa \geq 1$ ,  $\sum_{a \leq A} \sum_{m \leq B} \rho_{\kappa a}(m)/m \ll A \log B \log(1+\kappa)$ .

We use (a) and (d) only with  $\kappa = 4$ . Elsholtz and Tao state (d) inside the case  $A \leq B$  of their proof of (a), and we use it only in that range.

**Lemma 3.2** (Prime-power counts; proved). (a)  $h(xy) \leq h(x) + h(y)$  for all  $x, y \geq 1$ . (b) For  $Y \geq 2$  and  $N \geq 1$ ,  $\sum_{q|N, q > Y} 1/i \leq \log N / \log Y$ , where  $q = \ell^i$ . (c)  $h(M) \leq \log M / \log 3$  for odd  $M$ .

*Proof.* (a)  $h = \sum_{\ell} H_{v_{\ell}}$ , and  $H_{x+y} - H_x = \sum_{x < i \leq x+y} 1/i \leq H_y$ . (b) Fix  $\ell \mid N$  with  $v = v_{\ell}(N)$ . Each  $i \leq v$  with  $\ell^i > Y$  has  $1/i < \log \ell / \log Y$ , so these terms contribute at most  $v \log \ell / \log Y$ . Sum over  $\ell$ . (c)  $h(M) \leq \sum_{\ell} v_{\ell}(M)$ .  $\square$

**Lemma 3.3** (Parametrisation; proved, cf. [10, §2]). Let  $(M, D)$  be an atom with  $D \leq A_M$ . There are unique integers  $a, b, d \geq 1$  with  $d$  squarefree,  $D = da^2$ ,  $A_M = dab$  and  $a \leq b$ . Put  $P = 4a^2d + 1 = 4D + 1$ ,  $e = g(M, D)$ ,  $f = P/e$ ,  $c = (a+b)/e$  and  $N = M/e$ . Then  $c$  is a positive integer,  $f \mid P$ , and

$$N = 4acd - f, \quad acd \leq N \leq T, \quad a^2d \leq T, \quad \frac{g}{M} = \frac{1}{N}.$$

The map  $(M, D) \mapsto (a, c, d, f)$  is injective. The involution  $D \mapsto A_M^2/D$  of the divisors of  $A_M^2$  preserves  $g(M, D)$ .

*Proof.* Write  $D = da^2$  with  $d$  squarefree. Comparing valuations,  $da^2 \mid A_M^2$  if and only if  $da \mid A_M$ ; so  $A_M = dab$ , and  $D \leq A_M$  if and only if  $a \leq b$ . As  $M = 4dab - 1$ ,  $\gcd(M, 4dab) = 1$ . Since  $M + 4D + 1 = 4da(a+b)$ ,  $e \mid 4da(a+b)$  and hence  $e \mid a+b$ , so  $c$  is an integer;  $e \mid P$  by definition. Next,  $e \cdot 4acd = 4ad(a+b) = (P-1) + (M+1) = ef + eN$ , so  $4acd = f + N$ . From  $b \geq a$ ,  $eN = 4abd - 1 \geq 4a^2d - 1 = ef - 2$ , so  $N \geq f - 2/e$ ; then  $2N \geq N + f - 2/e = 4acd - 2/e$ , and  $N \geq 2acd - 1 \geq acd$ . Also  $a^2d = D \leq A_M \leq T$ ,  $N \leq M \leq T$  and  $g/M = e/(eN)$ . The quadruple determines

$e = P/f$ , then  $b = ce - a$ ,  $D$  and  $A_M$ , hence the atom. For the involution:  $\gcd(A_M, M) = 1$  and  $4A_M \equiv 1 \pmod{M}$ , so  $4A_M^2/D + 1 \equiv (4D)^{-1}(1 + 4D) \pmod{M}$ , and  $g$  is unchanged.  $\square$

The involution exchanges  $D < A_M$  with  $D > A_M$  and preserves  $M$ , so

$$S_0 \leq 2 \sum_{(a,c,d,f)} \frac{1}{N}, \quad \Omega_0 \leq 2 \sum_{(a,c,d,f)} \frac{h(eN)}{N} \leq 2(\Sigma_I + \Sigma_{II}), \quad (3)$$

where  $\Sigma_I = \sum h(e)/N$  and  $\Sigma_{II} = \sum h(N)/N$ , with all sums over the quadruples of Lemma 3.3 and  $e = P/f$ ; we used Lemma 3.2(a). For dyadic  $A, B, C \in \{2^j : j \geq 0\}$  the *block*  $(A, C, B)$  consists of the quadruples with  $a \in [A, 2A)$ ,  $c \in [C, 2C)$ ,  $d \in [B, 2B)$ . Only blocks with  $A^2B \leq T$  and  $ACB \leq T$  occur. There are  $O(\mathcal{L}^2)$  pairs  $(A, B)$  and  $O(\mathcal{L})$  values of  $C$ .

**Lemma 3.4** (Block mass; proved modulo Theorem 3.1(a)). *For every block,*

$$\sum_{(a,c,d,f) \in (A,C,B)} \frac{1}{N} \leq \sum_{a \in [A, 2A), d \in [B, 2B)} \frac{2\tau(4a^2d + 1)}{ad} \ll \log(A + B + 2).$$

Consequently  $S_0 \ll \mathcal{L}^4$ . Unconditionally,  $S_0 \leq 2(1 + \mathcal{L})^3 \tau^*(4T + 1)$ .

*Proof.*  $1/N \leq 1/(acd)$ ,  $\sum_{c \in [C, 2C)} 1/c \leq 1 + \log 2 < 2$ , and for fixed  $(a, d)$  there are  $\tau(P)$  choices of  $f$ . In the block  $1/(ad) \leq 1/(AB)$ , and Theorem 3.1(a) with  $\kappa = 4$  and Elsholtz–Tao’s  $(a, b, A, B)$  equal to our  $(d, a, 2B, 2A)$  gives  $\sum_{a < 2A, d < 2B} \tau(4a^2d + 1) \ll AB \log(A + B + 2)$ . Sum over the  $O(\mathcal{L}^3)$  blocks, using  $\log(A + B + 2) \ll \mathcal{L}$ . For the unconditional bound, use  $\tau(P) \leq \tau^*(4T + 1)$  and  $\sum_{a,c,d \leq T} 1/(acd) \leq (1 + \mathcal{L})^3$  in (3).  $\square$

**Lemma 3.5** (The rough part; proved modulo Theorem 3.1(a)). *For every block  $(A, C, B)$ , with  $Y = \max(C, 2)$ ,*

$$\sum_{(a,c,d,f) \in (A,C,B)} \frac{h(N)}{N} \ll \log(A + B + 2) \left( \log \log(C + 2) + 1 + \frac{\mathcal{L}}{\log Y} \right).$$

Hence  $\Sigma_{II} \ll \mathcal{L}^4 \log \mathcal{L}$ .

*Proof.*  $N$  divides the odd number  $M$ , so only odd  $q = \ell^i$  divide  $N$ . Write  $h(N) = \sum_{q|N} 1/i$  and split at  $q \leq C$  and  $q > Y$ ; for  $C = 1$  the first part is empty.

*Small  $q$ .* Fix  $(a, d, f)$  and  $q = \ell^i \leq C$ . If  $\ell \mid ad$ , then  $P \equiv 1 \pmod{\ell}$ , so  $\ell \nmid f$  and  $N \equiv -f \not\equiv 0 \pmod{\ell}$ . Otherwise  $q \mid N = 4adc - f$  confines  $c$  to one class modulo  $q$ , which has at most  $C/q + 1 \leq 2C/q$  elements in  $[C, 2C)$ , and each term has  $1/N \leq 1/(adC)$ . So the contribution of  $(a, d, f)$  is at most  $(2/(ad)) \sum_{q \leq C} 1/(iq) \leq (2/(ad))(\log \log(C + 2) + O(1))$  by Mertens’ theorem. Summing over the  $\tau(P)$  choices of  $f$  and over  $(a, d)$  as in Lemma 3.4 gives  $\ll \log(A + B + 2)(\log \log(C + 2) + 1)$ .

*Large  $q$ .* By Lemma 3.2(b) and  $N \leq T$ ,  $\sum_{q|N, q > Y} 1/i \leq \mathcal{L}/\log Y$  pointwise, and Lemma 3.4 bounds the mass of the block.

*Sum.* With  $C = 2^j$ ,  $0 \leq j \leq \mathcal{L}/\log 2$ ,

$$\sum_j \left( \log \log(2^j + 2) + 1 + \frac{\mathcal{L}}{\log \max(2^j, 2)} \right) \ll \mathcal{L} \log \mathcal{L},$$

and  $\sum_{(A,B)} \log(A + B + 2) \ll \mathcal{L}^3$ .  $\square$

The prime powers  $q \leq C$  are summed along the  $c$ -progression only where a block contains a full period. The larger ones are paid pointwise, at the cost  $\mathcal{L}/\log C$ ; this is large only for small  $c$ , and every dyadic  $c$ -scale carries the same mass bound, so the cost averages to  $\mathcal{L}^{-1} \sum_{j \leq \mathcal{L}} \mathcal{L}/j \asymp \log \mathcal{L}$ .

For the smooth part note that  $\sum_{e|P} h(e) = \sum_{q|P} (1/i) \tau(P/q)$  (write  $e = qe'$  with  $e' \mid P/q$ ). For a pair  $(A, B)$  put  $Z_0 = \max(A, B)$  and

$$R(A, B) = \sum_{a \in [A, 2A), d \in [B, 2B)} \frac{1}{ad} \sum_{q|P} \frac{1}{i} \tau(P/q), \quad P = 4a^2d + 1.$$

**Lemma 3.6** (Prime powers of  $4a^2d + 1$ ; proved modulo Theorem 3.1).  $R(A, B) \ll \log(Z_0 + 2) \log \log(Z_0 + 16)$ , uniformly in  $A, B \geq 1$ . Hence  $\Sigma_I \ll \mathcal{L}^4 \log \mathcal{L}$ .

*Proof.* Every  $q = \ell^i \mid P$  is odd and has  $\ell \nmid ad$ , since  $P$  is odd and  $P \equiv 1 \pmod{\ell}$  if  $\ell \mid ad$ .

*Large  $q$*  ( $q > Z_0^{1/2}$ ), and all  $q$  if  $Z_0 < 16$ . Use  $\tau(P/q) \leq \tau(P)$ . As  $P < 32Z_0^3$ , Lemma 3.2(b) gives  $\sum_{q|P, q > Z_0^{1/2}} 1/i \leq 2 \log(32Z_0^3)/\log Z_0 \ll 1$  for  $Z_0 \geq 16$ , and for  $Z_0 < 16$ ,  $h(P) \leq \log_2 P \ll 1$ . So this part is  $\ll \sum \tau(P)/(ad) \ll \log(Z_0 + 2)$  by Theorem 3.1(a), as in Lemma 3.4.

*Small  $q$ , case  $A \geq B$*  (so  $Z_0 = A \geq 16$  and  $q \leq A^{1/2}$ ). Fix  $d < 2B$  with  $\ell \nmid d$ . The congruence  $4dx^2 + 1 \equiv 0 \pmod{q}$  has at most two roots  $x_0 \in [0, q)$  ( $\ell$  odd,  $\ell \nmid 4dx_0$ , Hensel). For each, write  $a = qa' + x_0$ ; then  $1 \leq a' \leq N' := \lfloor 2A/q \rfloor$ , as  $a \geq A \geq q > x_0$ . Put

$$P^\flat(a') = \frac{P(qa' + x_0)}{q} = 4dq a'^2 + 8dx_0 a' + \frac{4dx_0^2 + 1}{q}.$$

Its coefficients are non-negative integers at most  $8dq \leq 16A^{3/2} \leq N'^5$  (as  $N' \geq A^{1/2}$  and  $A \geq 16$ ). It takes odd values, so  $\rho_{P^\flat}(2^j) = 0$ . For an odd prime  $p \neq \ell$ ,  $a' \mapsto qa' + x_0$  is a bijection modulo  $p^j$ , so  $\rho_{P^\flat}(p^j) = \rho_{4d}(p^j) \leq 2$ . For  $p = \ell$ , each of the at most two roots of  $4dx^2 + 1$  modulo  $\ell^{i+j}$  fixes  $a'$  modulo  $\ell^j$ , so  $\rho_{P^\flat}(\ell^j) \leq 2$ . By Theorem 3.1(b) (degree 2,  $l = 5$ ,  $C = 2$ ) and multiplicativity,

$$\sum_{a' \leq N'} \tau(P^\flat(a')) \ll N' \sum_{m \leq N'} \frac{\rho_{P^\flat}(m)}{m} \leq \frac{2A}{q} \left( 1 + \frac{2}{\ell - 1} \right) \sum_{m \leq 2A} \frac{\rho_{4d}(m)}{m}.$$

With  $1/(ad) \leq 1/(AB)$  and at most two roots,

$$\sum_{a, d: q|P} \frac{\tau(P/q)}{ad} \ll \frac{1}{qB} \sum_{d < 2B} \sum_{m \leq 2A} \frac{\rho_{4d}(m)}{m} \ll \frac{\log A}{q},$$

by Theorem 3.1(d) with  $\kappa = 4$  and Elsholtz–Tao’s  $(A, B)$  equal to our  $(2B, 2A)$ , which satisfies  $2 \leq 2B \leq 2A$ .

*Small  $q$ , case  $B > A$*  (so  $Z_0 = B \geq 16$  and  $q \leq B^{1/2}$ ). Fix  $a < 2A$  with  $\ell \nmid a$ . Then  $q \mid P$  means  $d \equiv d_0 \pmod{q}$ , where  $d_0 \equiv -(4a^2)^{-1}$  and  $0 < d_0 < q$ . Write  $d = qd' + d_0$  with  $1 \leq d' \leq N'' := \lfloor 2B/q \rfloor$  (as  $d \geq B \geq q$ ). Then  $P/q = 4a^2d' + b_a$  with  $b_a = (4a^2d_0 + 1)/q \leq 4a^2$ , and  $\gcd(4a^2, b_a) = 1$ :  $b_a$  is odd, and a prime dividing  $a$  and  $b_a$  divides  $4a^2d_0 + 1$ . Both coefficients are at most  $16B^2 \leq N''^6$ . By Theorem 3.1(c),  $\sum_{d' \leq N''} \tau(4a^2d' + b_a) \ll N'' \log N'' \leq (2B/q) \log(2B)$ . With  $1/(ad) \leq 1/(aB)$  we get  $\sum_{a,d: q \mid P} \tau(P/q)/(ad) \ll (\log B/q) \sum_{a \in [A, 2A]} 1/a \ll \log B/q$ .

*Sum over small  $q$ .*  $\sum_{q \leq Z_0^{1/2}} (1/q) \log Z_0/q \ll \log Z_0 \log \log Z_0$ .

$\Sigma_I$ . We have  $1/N \leq 1/(acd)$  and  $\sum_{c \in [C, 2C]} 1/c < 2$ , and each  $(a, d, e)$  with  $e \mid P$  occurs at most once for each  $c$ . So  $\Sigma_I \leq \sum_C \sum_{(A, B)} 2R(A, B) \ll \mathcal{L} \cdot \mathcal{L}^2 \cdot \mathcal{L} \log \mathcal{L}$ , as  $Z_0 \leq T$ .  $\square$

**Theorem 3.7** (Charge moment; proved modulo Theorem 3.1).  $S_0 \ll \mathcal{L}^4$  and  $\Omega_0 \ll \mathcal{L}^4 \log \mathcal{L}$ . Unconditionally,  $\Omega_0 \leq \mathcal{L} S_0$  and  $\log S_0 \leq (\log 2 + o(1)) \mathcal{L} / \log \mathcal{L}$ .

*Proof.* Lemma 3.4, (3) and Lemmas 3.5 and 3.6. Unconditionally,  $h(M) \leq \mathcal{L}$  by Lemma 3.2(c), and  $\log \tau^*(x) \leq (\log 2 + o(1)) \log x / \log \log x$  [17, Thm. 317].  $\square$

**Remark 3.8** (Assessment). Theorem 3.7 bounds the moment, not the  $g/M$ -weighted mean  $\Omega_0/S_0$  of  $h(M)$ ; that would need  $S_0 \gg \mathcal{L}^4$ , which we have not proved. Numerically (atoms with  $D \leq A_M$ ,  $T = 10^4, 10^5, 10^6$ ) the mean is 2.16, 2.40, 2.62, which tracks  $\log \log T$ , and the splitting  $h(eN) \leq h(e) + h(N)$  loses about 2%. This is evidence only.

#### 4. THE ANALYTIC INPUT: GALLAGHER’S THEOREM

For a Dirichlet character  $\chi$  put  $\vartheta(x; \chi) = \sum_{p \leq x} \chi(p) \log p$ , and let  $E_0(\chi) = 1$  if  $\chi$  is the trivial character modulo 1 and  $E_0(\chi) = 0$  otherwise.  $\sum_{\chi \bmod q}^*$  is a sum over primitive characters.

**Theorem 4.1** (Landau–Page; classical, [7, Ch. 14], [21, Cor. 11.10]). *There are absolute effective positive constants  $c_1, c_2$  such that, for every  $Y \geq 3$ ,  $\prod_{q \leq Y} \prod_{\chi \bmod q}^* L(s, \chi)$  has at most one zero in the region  $\Re s > 1 - c_1/\log Y$ ,  $|\Im s| \leq Y$ . If this zero  $\beta_1$  exists, it is real and simple, and its character  $\chi_1$  is quadratic. Moreover, every real zero  $\beta$  of  $L(s, \chi)$ , for  $\chi$  a real primitive character of conductor  $q \geq 3$ , satisfies  $1 - \beta \geq c_2^{-1} q^{-1/2} (\log q)^{-2}$ .*

The first part is the classical zero-free region with the Landau–Page theorem. The height restriction matters: without it the statement would be a zero-free region of width  $\asymp 1/\log Y$  uniformly in  $|\Im s|$ , which is not known. The “Exceptional Zero Statement” (28.61)–(28.62) of [22], which refers to [21, Cor. 11.10], omits the height restriction. The second part is the effective

Page-type bound [7, Ch. 14], [22, (28.62)]. We use Theorem 4.1 only for *real* zeros: for the uniqueness of a real zero close to 1, for the quadraticity of its character, and for the lower bound for  $1 - \beta$ .

**Theorem 4.2** (Gallagher [12]; cited in the form of [22, Thm. 28.19]). *There are absolute constants  $c \geq 1$  and  $\kappa_0 \geq 3$  with the following property. Let  $\kappa \geq \kappa_0$  be fixed and  $1 < Q_G^{6c} \leq x$ . Then*

$$\sum_{q \leq Q_G} \sum_{\chi \bmod q}^* |\vartheta(x; \chi) - E_0(\chi)x| \ll_{\kappa} x \exp\left(-\frac{\log x}{\kappa \log Q_G}\right) + \left(\frac{\log x}{\log Q_G}\right)^2 \frac{x}{Q_G},$$

*unless  $\prod_{q \leq Q_G} \prod_{\chi \bmod q}^* L(s, \chi)$  has a real zero  $\beta_1$  with  $1 - \beta_1 < 1/(\kappa \log Q_G)$ . In that case the term of the exceptional character  $\chi_1$  on the left is replaced by  $|\vartheta(x; \chi_1) + x^{\beta_1}/\beta_1|$ , and the right-hand side by*

$$(1 - \beta_1)(\log x) \left( x \exp\left(-\frac{\log x}{\log Q_G}\right) + \frac{x \log x}{Q_G \log Q_G} \right).$$

*Source caveat.* We quote [22], an unpublished draft (third volume of Montgomery–Vaughan), archived by us. We have not seen Gallagher’s paper [12], so we cite it without a theorem number. The printed proof of [22, Thm. 28.19] has visible slips. Its case split is at  $1 - \beta_1 < 1/(3\kappa_0 \log Q_G)$ , so it establishes the statement for  $\kappa = 3\kappa_0$ . The statement for every larger  $\kappa$  follows from that case. Let  $\kappa \geq 3\kappa_0$ , and suppose a real zero satisfies  $1/(\kappa \log Q_G) \leq 1 - \beta_1 < 1/(3\kappa_0 \log Q_G)$ . Then  $x^{\beta_1}/\beta_1 \leq 2x \exp(-\log x/(\kappa \log Q_G))$ , and  $(1 - \beta_1)(\log x) x \exp(-\log x/\log Q_G) \ll x \exp(-\log x/(2 \log Q_G))$ . So the exceptional form of the bound implies the non-exceptional one with parameter  $\kappa$ . We therefore fix

$$\kappa := \max(3\kappa_0, 1/c_1).$$

An exceptional zero as in Theorem 4.2 is then a real zero in the region of Theorem 4.1 with  $Y = Q_G$ , so it is unique and  $\chi_1$  is real. Write  $C_G$  for the implied constant in Theorem 4.2 with this  $\kappa$ .

Gallagher’s original statement may carry a range condition of the form  $\exp(\sqrt{\log x}) \leq Q_G \leq x^b$ ; we have not checked this. Our use is robust to such a condition. In the proof of Theorem 5.1,  $\log x/\log Q_G = \kappa L \geq 6c$ , and  $\log x \geq (\kappa L)^2$  by the hypothesis of that theorem, so  $Q_G \geq \exp(\sqrt{\log x})$ . The statement itself is the standard form of Gallagher’s prime number theorem for the whole family of characters of conductor  $\leq Q_G$ , which underlies the usual proofs of Linnik’s theorem.

## 5. A LINEAR TRANSFER THEOREM

**Theorem 5.1** (Linear transfer; proved modulo Theorems 4.1 and 4.2). *Let  $T \geq 1$  and  $Q \geq 2$  with  $8 \mid Q$ . Let  $B(n) = \sum_{i \in I} c_i \mathbf{1}[n \equiv b_i \pmod{d_i}]$  be a finite real combination of consistent unit cells:  $\gcd(b_i, d_i) = 1$  and  $b_i \equiv 1 \pmod{\gcd(d_i, Q)}$ . Assume that  $B(n) \leq \mathbf{1}[W(n) > T]$  for every*

$n \equiv 1 \pmod{Q}$  coprime to all  $d_i$ . Put  $D = \text{lcm}_i d_i$ ,  $N_0 = \text{lcm}(Q, D)$ ,  $H = \{n \in (\mathbb{Z}/N_0)^\times : n \equiv 1 \pmod{Q}\}$ , and let  $\mathbb{E}_H$  be the mean over  $H$ . Put

$$Z = Q \max_i d_i, \quad \mu = \mathbb{E}_H B, \quad A = \frac{\mathbb{E}_H |B|}{\mu},$$

and assume  $\mu > 0$ ,  $A \leq Z^{1/4}$ , and the twist condition: for every real primitive character  $\psi$  of conductor  $f > 1$  with  $\gcd(f, Q) = 1$  and  $f \mid d_i$  for some  $i$ ,

$$|\mathbb{E}_H [B\psi]| \leq \mu/4.$$

There is an absolute effective constant  $C_2$  such that, if

$$\log x \geq C_2(1 + \log A) \log Z,$$

then some prime  $p \leq x$  with  $p \equiv 1 \pmod{Q}$  and  $p \nmid D$  has  $W(p) > T$ .

In the previous version the cells were coprime to  $Q$ ; then  $\mathbb{E}_H$  is the mean over the units modulo  $D$ , and  $\mu = \sum_i c_i/\varphi(d_i)$ . Graded quarantines produce cells whose moduli share primes with  $Q$ ; consistency means that each cell meets the class 1 mod  $Q$ .

Neither the  $\ell^1$  mass  $\sum_i |c_i|/\varphi(d_i)$  nor the number of cells enters; only the Haar quantity  $A \geq 1$  does.

*Proof. Parameters.* Put  $L = 2c + \log(400C_G(A+1))$  and  $Q_G = x^{1/(\kappa L)}$ . Since  $A \geq 1$  we have  $L \leq C_3(1 + \log A)$  for an absolute  $C_3$ . If  $C_2$  is large enough, the hypothesis gives the following, using  $A \leq Z^{1/4}$  and  $Z \geq 2$ :  $\log Q_G \geq (C_2/(\kappa C_3)) \log Z$ , hence  $Q_G \geq Z$  and  $Q_G \geq 10^4 C_G(A+1)(\kappa L)^2$ ;  $Q_G^{6c} \leq x$  (as  $\kappa L \geq 6c$ );  $\log x \geq 16$ ;  $\log x \geq (\kappa L)^2$  (as  $(1 + \log A)^2 \leq (1 + \log A)(1 + \frac{1}{4} \log Z)$ ); and  $x \geq C_4 A Z^4$ , with  $C_4$  as large as needed below.

*Character expansion.* The function  $f(n) = B(n)\mathbf{1}[n \equiv 1 \pmod{Q}]$  on units modulo  $N_0$  satisfies  $f(n) = \sum_{\chi \bmod N_0} c(\chi)\chi(n)$ . The reduction map  $(\mathbb{Z}/N_0)^\times \rightarrow (\mathbb{Z}/Q)^\times$  is onto with kernel  $H$ , so  $|H| = \varphi(N_0)/\varphi(Q)$  and

$$c(\chi) = \frac{1}{\varphi(N_0)} \sum_{n \in H} B(n)\bar{\chi}(n) = \frac{\mathbb{E}_H [B\bar{\chi}]}{\varphi(Q)}.$$

Consequently:

- (a)  $c(\chi) = 0$  unless the conductor of  $\chi$  divides  $\text{lcm}(d_i, Q)$  for some  $i$ . Indeed, by consistency and the Chinese remainder theorem,  $\mathbf{1}[n \equiv b_i \pmod{d_i}]\mathbf{1}[n \equiv 1 \pmod{Q}]$  is the indicator of one unit class modulo  $\text{lcm}(d_i, Q)$ , and its  $\chi$ -coefficient vanishes unless  $\chi$  is trivial on  $\{n \equiv 1 \pmod{\text{lcm}(d_i, Q)}\}$ . So  $c(\chi) \neq 0$  implies that the conductor of  $\chi$  is at most  $Qd_i \leq Z \leq Q_G$ .
- (b)  $|c(\chi)| \leq \mathbb{E}_H |B|/\varphi(Q) = A\mu/\varphi(Q)$ , and  $c(\chi_0) = \mu/\varphi(Q)$  for the principal character  $\chi_0$ .
- (c) Let  $\chi$  be real with  $c(\chi) \neq 0$ , and let  $\psi^*$  be the real primitive character inducing it, of conductor  $f$ . Write  $f = f_1 f_2$ , where  $f_1$  is composed of primes dividing  $Q$  and  $\gcd(f_2, Q) = 1$ , and  $\psi^* = \psi_1 \psi_2$  accordingly. A real primitive conductor is  $2^e f'$  with  $e \in \{0, 2, 3\}$  and  $f'$  odd and

squarefree; as  $8 \mid Q$ ,  $f_1 \mid Q$ , so  $\psi_1 = 1$  on  $H$  and  $\chi = \psi_2$  on  $H$ . If  $f_2 = 1$ , then  $\chi$  is trivial on  $H$ , so it is induced from a character modulo  $Q$  and  $c(\chi) = \mu/\varphi(Q)$ . If  $f_2 > 1$ , then  $c(\chi) = \mathbb{E}_H[B\psi_2]/\varphi(Q)$ . A cell with  $f_2 \nmid d_i$  has  $\mathbb{E}_H[\mathbf{1}[n \equiv b_i \pmod{d_i}]\psi_2] = 0$ : there is a prime  $p \mid f_2$  with  $p \nmid d_i Q$ ; on  $H$  the residue of  $n$  modulo  $p^{v_p(N_0)}$  is uniform among units and independent of  $n$  modulo  $N_0/p^{v_p(N_0)}$ , and the  $p$ -component of  $\psi_2$  is a nontrivial character modulo  $p$ . So  $f_2 \mid d_i$  for some  $i$ ,  $\gcd(f_2, Q) = 1$ , and the twist condition gives  $|c(\chi)| \leq \mu/(4\varphi(Q))$ .

Let  $\chi^*$  be the primitive character inducing  $\chi$ . For a fixed modulus the map  $\chi \mapsto \chi^*$  is injective, so by (a) at most  $\sum_{r \leq Z} \varphi(r) \leq Z^2$  characters have  $c(\chi) \neq 0$ .

*The prime sum.* Let  $S(x) = \sum_{p \leq x, p \nmid N_0} f(p) \log p = \sum_{\chi} c(\chi) \vartheta(x; \chi)$ . Since  $|\vartheta(x; \chi) - \vartheta(x; \chi^*)| \leq \log N_0 \leq \log Q + \log D \leq 2Z$  (as  $D \mid \text{lcm}(1, \dots, \max d_i)$ ), so  $\log D \leq \psi(\max d_i) < 1.04 \max d_i$  by Rosser–Schoenfeld [30]),

$$S(x) = \frac{\mu x}{\varphi(Q)} + \sum_{\chi: c(\chi) \neq 0} c(\chi) (\vartheta(x; \chi^*) - E_0(\chi^*)x) + R_1, \quad (4)$$

where  $|R_1| \leq Z^2 \cdot (A\mu/\varphi(Q)) \cdot 2Z = 2AZ^3\mu/\varphi(Q)$ . Here  $E_0(\chi^*) = 1$  only for  $\chi = \chi_0$ , and  $|R_1| \leq \mu x/(400\varphi(Q))$  as  $x \geq 800AZ^3$ .

*Case 0: no exceptional zero* in the sense of Theorem 4.2. The  $\chi^*$  in (4) are distinct primitive characters of conductor  $\leq Q_G$ . By (b) and Theorem 4.2, the middle sum is at most

$$\frac{A\mu}{\varphi(Q)} C_G x \left( e^{-L} + \frac{(\kappa L)^2}{Q_G} \right) \leq \frac{\mu x}{\varphi(Q)} \left( \frac{1}{400} + \frac{1}{10^4} \right) \leq \frac{\mu x}{200\varphi(Q)},$$

using  $\log x / \log Q_G = \kappa L \geq L$ . So  $S(x) > 0$ .

*Exceptional case:* a zero  $\beta_1$  with character  $\chi_1$  and  $1 - \beta_1 < 1/(\kappa \log Q_G)$ . Put  $u = (1 - \beta_1) \log x$ ; then  $u < \log x / (\kappa \log Q_G) = L$ , and  $\beta_1 > 1/2$ . The replaced right-hand side of Theorem 4.2 is  $C_G u x (e^{-\kappa L} + \kappa L / Q_G)$ . Since  $u \leq \min(u, 1) L$  (recall  $L \geq 2$ ), its product with  $A\mu/\varphi(Q)$  is at most

$$\frac{\mu x}{\varphi(Q)} \min(u, 1) \cdot C_G A L \left( e^{-\kappa L} + \frac{\kappa L}{Q_G} \right) \leq \frac{\mu x}{200\varphi(Q)} \min(u, 1).$$

Here  $C_G A L e^{-3L} \leq C_G A e^{-L} \leq 1/400$ , and  $C_G A \kappa L^2 / Q_G \leq 10^{-4}$ . If no  $\chi$  with  $c(\chi) \neq 0$  has  $\chi^* = \chi_1$ , this bounds the middle sum of (4), and  $S(x) > 0$  as in Case 0. Otherwise, by injectivity, exactly one such  $\chi$  exists. It is real, because  $\chi_1$  is; hence  $c(\chi)$  is real, as  $B$  and  $\chi$  are. Let  $f_2$  be as in (c). Its term is  $c(\chi)(\vartheta(x; \chi_1) + x^{\beta_1}/\beta_1) - c(\chi)x^{\beta_1}/\beta_1$ . The first part is covered by the bound just proved.

*Case A:*  $f_2 = 1$ . Then  $c(\chi) = \mu/\varphi(Q)$ , and the conductor  $q_1$  of  $\chi_1$  divides  $Q$ , so  $q_1 \leq Z$ . The main terms combine to  $\lambda \mu x / \varphi(Q)$  with  $\lambda = 1 - x^{\beta_1-1}/\beta_1$ . Since  $1/\beta_1 \leq 1 + 2(1 - \beta_1)$ ,  $1 - e^{-u} \geq (1 - e^{-1}) \min(u, 1)$  and  $2(1 - \beta_1)e^{-u} = 2ue^{-u}/\log x \leq 2 \min(u, 1)/\log x$ ,

$$\lambda \geq 1 - e^{-u} - 2(1 - \beta_1)e^{-u} \geq (0.63 - 2/16) \min(u, 1) \geq \min(u, 1)/2.$$

So the error from Theorem 4.2 is at most  $\lambda\mu x/(100\varphi(Q))$ . By Theorem 4.1,  $u \geq 16(1 - \beta_1) \geq 16/(c_2 Z^{1/2}(\log Z)^2)$ , so  $1/\lambda \leq 2 + c_2 Z^{1/2}(\log Z)^2/8 \leq C_5 Z$ . Hence  $|R_1| \leq 2AZ^3\mu/\varphi(Q) \leq \lambda\mu x/(100\varphi(Q))$  once  $x \geq 200C_5AZ^4$ . Altogether  $S(x) \geq (1 - 2/100)\lambda\mu x/\varphi(Q) > 0$ .

*Case B:*  $f_2 > 1$ . By (c),  $|c(\chi)x^{\beta_1}/\beta_1| \leq (\mu/(4\varphi(Q))) \cdot 2x = \mu x/(2\varphi(Q))$ , and  $S(x) \geq (1 - 1/2 - 1/200 - 1/400)\mu x/\varphi(Q) > 0$ .

In all cases  $S(x) > 0$ . So some prime  $p \leq x$  with  $p \nmid N_0$  and  $p \equiv 1 \pmod{Q}$  has  $B(p) > 0$ , and then  $W(p) > T$ .  $\square$

**Remark 5.2.** This is Linnik's theorem with weights. Every character coefficient is bounded by  $\mathbb{E}_H|B|/\varphi(Q)$ , not by the  $\ell^1$  mass of a cell representation. Theorem 4.2 controls all characters of conductor  $\leq Q_G$  simultaneously, so the number of moduli  $d_i$  never multiplies the error. In [2] and in the first version of this note, each progression was instead treated separately with the Thorner–Zaman theorem [32]. That costs  $\log x \gg K \max(\log Z, K)$  with  $K = 1 + \log(\sum |c_i|/\varphi(d_i)/\mu)$ , which here is of order  $(\log T)^{14}$  instead of  $(\log T)^7$  in the setting of the previous version. No lower bound for  $1 - \beta_1$  is needed in Case B. In Case A we use the effective bound of Theorem 4.1 only to absorb the term  $R_1$ .

## 6. AN ENERGY BOUND FOR AVOIDANCE INDICATORS

This section is self-contained and does not use arithmetic.

*Setting.* Let  $\Omega = \prod_{v \in V_0} \Omega_v$  be a finite product of finite sets with a product probability measure  $\pi = \bigotimes_v \pi_v$ ; write  $\mathbb{P}$ ,  $\mathbb{E}$  and  $\|\cdot\| = \|\cdot\|_{L^2(\pi)}$ . A *cylinder event* is a set  $E = \{X_v = \sigma_v \mid (v \in \text{supp } E)\}$  with  $\text{supp } E \subseteq V_0$  and  $\sigma \in \prod_{\text{supp } E} \Omega_v$ . For a finite family  $\mathcal{E}$  of cylinder events (repetitions allowed) let  $F = F_{\mathcal{E}} = \prod_{E \in \mathcal{E}} (1 - \mathbf{1}_E)$  be its *avoidance indicator*.

*Efron–Stein decomposition* (classical; [8, 18], [26, §8.3]). For  $\varphi : \Omega \rightarrow \mathbb{R}$  and  $W \subseteq V_0$  let  $\mathbb{E}[\varphi \mid X_W]$  be the conditional expectation given the coordinates in  $W$ , and

$$\varphi^{=U} = \sum_{W \subseteq U} (-1)^{|U \setminus W|} \mathbb{E}[\varphi \mid X_W].$$

Then  $\varphi = \sum_U \varphi^{=U}$ , the components are mutually orthogonal,  $\varphi^{=U}$  depends only on  $X_U$ , and  $\mathbb{E}_u \varphi^{=U} = 0$  for  $u \in U$ , where  $\mathbb{E}_u$  averages over the coordinate  $u$ . Put  $L_u = I - \mathbb{E}_u$  and  $L_V = \prod_{u \in V} L_u$ ; then  $L_V \varphi = \sum_{U \subseteq V} \varphi^{=U}$ . For a down-closed family  $\mathcal{D}$  of subsets of  $V_0$ , the function  $\varphi^{\mathcal{D}} = \sum_{U \in \mathcal{D}} \varphi^{=U}$  is a linear combination of the  $\mathbb{E}[\varphi \mid X_W]$ ,  $W \in \mathcal{D}$  (by Möbius inversion), and

$$\mathbb{E}(\varphi - \varphi^{\mathcal{D}})^2 = \sum_{U \notin \mathcal{D}} \|\varphi^{=U}\|^2. \quad (5)$$

For  $\mathcal{D} = \{U : |U| \leq t\}$  this is the *energy above level  $t$* ,  $\text{En}(\varphi; t)$ ; it is the least  $\mathbb{E}(\varphi - g)^2$  over sums  $g$  of functions of at most  $t$  coordinates.

**Definition 6.1** (Energy generating function). For weights  $\lambda_v \geq 1$  put  $\mu_v = \lambda_v - 1$  and

$$G_\varphi(\lambda) = \sum_{U \subseteq V_0} \lambda^U \|\varphi^{=U}\|^2 = \sum_{V \subseteq V_0} \mu^V \|L_V \varphi\|^2.$$

The second form follows from the identity  $\lambda^U = \sum_{V \subseteq U} \mu^V$  and  $\|\varphi^{=U}\|^2 = \sum_{V \subseteq U} \|\varphi^{=V}\|^2$ . For every  $\tau$ ,

$$\sum_{U: \lambda^U > e^\tau} \|\varphi^{=U}\|^2 \leq e^{-\tau} G_\varphi(\lambda). \quad (6)$$

For a single event  $E$  with  $\pi_E = \mathbb{P}(E)$  one computes  $G_{1-\mathbf{1}_E} = 1 - 2\pi_E + \pi_E \prod_{v \in \text{supp } E} (\lambda_v - \mu_v \pi_v(\sigma_v))$ , so for small  $\pi_v(\sigma_v)$  the condition  $\lambda^{\text{supp } E} \leq 2$  below is necessary for  $G \leq 1$ .

*Hypergraphs.* A hypergraph  $\mathcal{H}$  is a finite set of subsets (*edges*) of  $V_0$ . For  $V \subseteq V_0$  put

$$N_{\mathcal{H}}(V) = \sum_{\mathcal{J} \subseteq \mathcal{H}: \bigcup_{J \in \mathcal{J}} (J \cap V) = V} (-1)^{|\mathcal{J}|}, \quad Q_\mu(\mathcal{H}) = \sum_{V \subseteq V_0} \mu^V N_{\mathcal{H}}(V)^2.$$

$N_{\mathcal{H}}(V)$  depends only on the traces  $J \cap V$ , and it is unchanged if edges are repeated or replaced by edges with the same trace: a class of  $m \geq 1$  edges with equal traces contributes  $\sum_{j \geq 1} \binom{m}{j} (-1)^j = -1$ , exactly like one edge. If some edge misses  $V$ , then  $N_{\mathcal{H}}(V) = 0$  (toggle that edge).

**Lemma 6.2** (Cover bound; proved). *In the Setting, for every  $V \subseteq V_0$ ,*

$$\|L_V F\|^2 \leq \mathbb{E}_x [N_{\mathcal{H}(x)}(V)^2], \quad \mathcal{H}(x) = \{\text{supp } E : E \in \mathcal{E}, x \in E\}.$$

*Consequently  $G_F(\lambda) \leq \mathbb{E}_x [Q_\mu(\mathcal{H}(x))]$  for all weights  $\lambda_v \geq 1$ .*

*Proof.* Write  $x = (\sigma, y)$  with  $\sigma \in \Omega_V$ ,  $y \in \Omega_{V_0 \setminus V}$ . The operator  $L_V$  acts on the  $V$ -coordinates only, so  $\|L_V F\|^2 = \mathbb{E}_y \|L_V g_y\|_{L^2(\Omega_V)}^2$ , where  $g_y = F(\cdot, y)$ . Now  $g_y$  is the avoidance indicator on  $\Omega_V$  of the restricted events: those  $E \in \mathcal{E}$  whose conditions off  $V$  hold at  $y$ , restricted to  $\text{supp } E \cap V$ . Expand  $g_y = \sum_{\mathcal{J}} (-1)^{|\mathcal{J}|} \mathbf{1}_{\mathcal{J}}$  over subfamilies  $\mathcal{J}$  of restricted events, where  $\mathbf{1}_{\mathcal{J}}$  is the indicator of their intersection, a cylinder on  $\Omega_V$  (or empty).  $L_V$  kills every function that does not depend on all coordinates of  $V$ , hence every cylinder whose support is a proper subset of  $V$ . A cylinder with support  $V$  is a point indicator  $\mathbf{1}_{\{\sigma\}}$ . So  $L_V g_y = L_V c_y$ , where

$$c_y(\sigma) = \sum_{\mathcal{J}: \text{supp } \mathbf{1}_{\mathcal{J}} = V, \mathbf{1}_{\mathcal{J}} = \mathbf{1}_{\{\sigma\}}} (-1)^{|\mathcal{J}|}.$$

As  $L_V$  is an orthogonal projection in  $L^2(\pi_V)$ ,  $\|L_V c_y\|^2 \leq \mathbb{E}_\sigma c_y(\sigma)^2$ . The subfamilies  $\mathcal{J}$  counted in  $c_y(\sigma)$  are exactly the subfamilies of events that hold at  $x = (\sigma, y)$  whose traces on  $V$  cover  $V$ . Two events that hold at  $x$  and have the same support are the same cylinder, so by the remark above  $c_y(\sigma) = N_{\mathcal{H}(x)}(V)$ . Taking  $\mathbb{E}_y$  gives the first claim. Multiply by  $\mu^V$  and sum over  $V$  for the second.  $\square$

The suppression is built in:  $N_{\mathcal{H}(x)}(V) = 0$  as soon as some event that holds at  $x$  misses  $V$ .

For a finite multiset  $\mathcal{C}$  of subsets of  $V_0$  (possibly empty ones) put

$$\Theta_\lambda(\mathcal{C}) = \sum_{\mathcal{J} \subseteq \mathcal{C}} (-1)^{|\mathcal{J}|} \lambda^{|\bigcup \mathcal{J}|} \quad (\Theta_\lambda(\emptyset) = 1).$$

**Lemma 6.3** (Polarization; proved). *Let  $\lambda_v \geq 1$ ,  $\mu_v = \lambda_v - 1$ , and let  $P \subseteq V_0$  be random with  $\mathbb{P}(v \in P) = \mu_v/\lambda_v$  independently. For every hypergraph  $\mathcal{H}$ ,*

$$Q_\mu(\mathcal{H}) = \mathbb{E}_P[\Theta_\lambda(\mathcal{H}_P)^2], \quad \mathcal{H}_P = \{J \in \mathcal{H} : J \cap P = \emptyset\}.$$

*Proof.* Let  $\tau(R) = \mathbf{1}[R \cap J \neq \emptyset \ \forall J \in \mathcal{H}]$  for  $R \subseteq V_0$ , and  $\hat{\tau}(V) = \sum_{R \subseteq V} (-1)^{|V \setminus R|} \tau(R)$ . Expanding  $\tau(R) = \sum_{\mathcal{J} \subseteq \mathcal{H}} (-1)^{|\mathcal{J}|} \mathbf{1}[R \cap \bigcup \mathcal{J} = \emptyset]$ , and using that the Möbius transform of  $R \mapsto \mathbf{1}[R \cap U = \emptyset]$  is  $V \mapsto (-1)^{|V|} \mathbf{1}[V \subseteq U]$ , we get  $\hat{\tau}(V) = (-1)^{|V|} N_{\mathcal{H}}(V)$ . Let

$$\mathcal{T}(p) = \sum_V \hat{\tau}(V) p^V = \sum_R \tau(R) \prod_{v \in R} p_v \prod_{v \notin R} (1 - p_v)$$

be the multilinear extension of  $\tau$ . If the  $p_v$  are independent with mean 0 and variance  $\mu_v$ , distinct monomials are orthogonal, and  $\mathbb{E}[\mathcal{T}(p)^2] = \sum_V \mu^V \hat{\tau}(V)^2 = Q_\mu(\mathcal{H})$ . Take  $p_v = 1$  with probability  $\mu_v/\lambda_v$  and  $p_v = -\mu_v$  otherwise (mean  $\mu_v/\lambda_v - \mu_v/\lambda_v = 0$ , variance  $\mu_v/\lambda_v + \mu_v^2/\lambda_v = \mu_v$ ), and  $P = \{v : p_v = 1\}$ . Only the  $R \supseteq P$  contribute; writing  $R = P \cup B$  with  $B \subseteq P^c$ ,

$$\mathcal{T}(p) = \sum_{B \subseteq P^c} (-\mu)^B \lambda^{P^c \setminus B} \tau(P \cup B).$$

Now  $\tau(P \cup B) = \sum_{\mathcal{J} \subseteq \mathcal{H}_P} (-1)^{|\mathcal{J}|} \mathbf{1}[B \cap \bigcup \mathcal{J} = \emptyset]$ , and for  $U \subseteq P^c$ ,

$$\sum_{B \subseteq P^c \setminus U} (-\mu)^B \lambda^{P^c \setminus B} = \lambda^U \prod_{v \in P^c \setminus U} (\lambda_v - \mu_v) = \lambda^U.$$

Hence  $\mathcal{T}(p) = \Theta_\lambda(\mathcal{H}_P)$ .  $\square$

**Lemma 6.4** (Matching bound; proved). *If every  $J \in \mathcal{C}$  has  $w_J := \lambda^J \leq 2$ , then for every matching  $\mathcal{M} \subseteq \mathcal{C}$  (pairwise disjoint members)*

$$|\Theta_\lambda(\mathcal{C})| \leq \prod_{J \in \mathcal{M}} (w_J - 1); \quad \text{in particular } |\Theta_\lambda(\mathcal{C})| \leq 1.$$

*Proof.* For  $v \in V_0$  let  $\mathcal{C} - v$  be the members not containing  $v$  and  $\mathcal{C}/v = \{J \setminus \{v\} : J \in \mathcal{C}\}$ . Splitting  $\mathcal{J}$  according to whether it contains a member through  $v$ ,

$$\Theta(\mathcal{C}) = \Theta(\mathcal{C} - v) + \lambda_v [\Theta(\mathcal{C}/v) - \Theta(\mathcal{C} - v)] = \lambda_v \Theta(\mathcal{C}/v) - \mu_v \Theta(\mathcal{C} - v).$$

Both families again have all weights at most 2. We induct on  $|\bigcup \mathcal{C}|$ . If  $\emptyset \in \mathcal{C}$ , then  $\Theta(\mathcal{C}) = 0$  (toggling the empty member pairs the terms), and the claim holds. If  $\mathcal{C} = \emptyset$ , then  $\Theta = 1$  and  $\mathcal{M} = \emptyset$ . This covers the base case  $\bigcup \mathcal{C} = \emptyset$ . If  $\mathcal{M} = \emptyset$  and  $\mathcal{C} \neq \emptyset$ , it suffices to prove the bound for  $\mathcal{M} = \{J\}$  with any  $J \in \mathcal{C}$ , as  $w_J - 1 \leq 1$ . So let  $J_0 \in \mathcal{M}$  be nonempty,  $v \in J_0$ , and  $\Pi' =$

$\prod_{J \in \mathcal{M} \setminus \{J_0\}} (w_J - 1)$ . In  $\mathcal{C}/v$  the family  $(\mathcal{M} \setminus \{J_0\}) \cup \{J_0 \setminus \{v\}\}$  is a matching, so  $|\Theta(\mathcal{C}/v)| \leq (w_{J_0}/\lambda_v - 1)\Pi'$ . In  $\mathcal{C} - v$  the family  $\mathcal{M} \setminus \{J_0\}$  is a matching, so  $|\Theta(\mathcal{C} - v)| \leq \Pi'$ . Therefore  $|\Theta(\mathcal{C})| \leq (w_{J_0} - \lambda_v)\Pi' + \mu_v\Pi' = (w_{J_0} - 1)\Pi'$ .  $\square$

**Theorem 6.5** (Energy bound; proved). (a) *If every edge of a hypergraph  $\mathcal{H}$  has  $\lambda^J \leq 2$ , then  $Q_\mu(\mathcal{H}) \leq \prod_{J \in \mathcal{M}} (\lambda^J - 1)$  for every matching  $\mathcal{M} \subseteq \mathcal{H}$ ; in particular  $Q_\mu(\mathcal{H}) \leq 1$ .* (b) *In the Setting, if  $\lambda_v \geq 1$  and  $\lambda^{\text{supp } E} \leq 2$  for every  $E \in \mathcal{E}$ , then*

$$G_F(\lambda) = \sum_U \lambda^U \|F^{=U}\|^2 \leq 1.$$

*Proof.* (a) By Lemma 6.3 and Lemma 6.4 applied to  $\mathcal{H}_P$  with the matching  $\mathcal{M} \cap \mathcal{H}_P$ ,  $Q_\mu(\mathcal{H}) \leq \mathbb{E}_P \prod_{J \in \mathcal{M} \cap \mathcal{H}_P} (\lambda^J - 1)^2$ . The members of  $\mathcal{M}$  are disjoint, so the events  $\{J \cap P = \emptyset\}$  are independent, with probability  $\prod_{v \in J} \lambda_v^{-1} = 1/w_J$ . Hence  $Q_\mu(\mathcal{H}) \leq \prod_{J \in \mathcal{M}} [(1 - 1/w_J) + (w_J - 1)^2/w_J] = \prod_{J \in \mathcal{M}} (w_J - 1)$ . (b) Lemma 6.2 and (a), since  $\mathcal{H}(x)$  consists of supports of events.  $\square$

**Corollary 6.6** (Energy tail; proved). *In the Setting, if every event of  $\mathcal{E}$  has  $|\text{supp } E| \leq k$  ( $k \geq 1$ ), then for  $f \in \{F, 1 - F\}$  and every integer  $t \geq 0$ ,*

$$\text{En}(f; t) = \sum_{|U| > t} \|f^{=U}\|^2 \leq 2^{-(t+1)/k}.$$

*Proof.* Theorem 6.5(b) with  $\lambda_v = 2^{1/k}$  and (6);  $(1 - F)^{=U} = -F^{=U}$  for  $U \neq \emptyset$ .  $\square$

**Remark 6.7** (Sharpness). Over general product spaces the base  $2^{-1/k}$  cannot be improved. Take  $m$  events on pairwise disjoint sets of  $k$  coordinates, each fixing values of probability  $p$ . Then  $F^{=U}$  is supported on unions of whole supports, and letting  $p \rightarrow 0$  with  $m$  chosen optimally one finds numerically  $\max_m \text{En}(F; t) 2^{(t+1)/k} \asymp (t/k)^{-1/2}$ . So at most a factor  $\sqrt{t/k}$  could be gained. The condition  $\lambda^{\text{supp } E} \leq 2$  in Theorem 6.5 is necessary, by the single-event formula above.

**Remark 6.8** (Boolean form and comparison). A width- $k$  DNF  $g : \{\pm 1\}^n \rightarrow \{\pm 1\}$  is  $1 - 2F$  for the family of its terms (a term with a literal set is a cylinder event). As  $g^{=U} = -2F^{=U}$  for  $U \neq \emptyset$ , Corollary 6.6 gives, for every product measure on  $\{\pm 1\}^n$  (Efron–Stein weights; Fourier weights for the uniform measure),

$$\sum_{|S| > t} \hat{g}(S)^2 \leq 4 \cdot 2^{-(t+1)/k},$$

i.e.  $g$  is  $\varepsilon$ -concentrated up to degree  $k \log_2(4/\varepsilon)$ . The switching-lemma route of Linial, Mansour and Nisan gives  $\sum_{|S| \geq t} \hat{g}(S)^2 \leq 2 \cdot 2^{-t/(20k)}$  [26, §4.4], [20]. Also  $\lambda^{|U|} \geq 1 + |U| \ln \lambda$  gives total influence  $\sum_U |U| \|F^{=U}\|^2 \leq (k/\ln 2) \mathbb{P}(F = 0)$ . The same bounds hold for  $q$ -ary DNFs with literals  $x_v \in S_v$  (split each term into single-value cylinders on the same support;  $F$  and the supports do not change). On the uniform Boolean cube the truth lies between  $3^{-t/k}$  (an

OR of disjoint  $k$ -parities) and  $2^{-t/k}$ . We have not found Theorem 6.5 in the literature; see the introduction for what we checked.

*Digit filtrations.* For the application we need weights that grow along a filtration of each coordinate. Suppose that each coordinate  $X_\ell$  ( $\ell \in \mathcal{P}$ , a finite index set) is a tuple of independent *digits*  $(\delta_{\ell,i})_{i_0(\ell) \leq i < f_\ell}$ , with arbitrary distributions, and that the coordinates are independent; the Efron–Stein decomposition  $F = \sum_U F^{=U}$  below is taken over the digits, so  $U$  is a set of pairs  $(\ell, i)$ . For such  $U$  write  $U_\ell = \{i : (\ell, i) \in U\}$ . We say that an event  $E$  is a *digit-prefix event* if, for some integers  $v_\ell(E) \geq 0$ , it fixes exactly the digits  $\delta_{\ell,i}$  with  $i_0(\ell) \leq i < v_\ell(E)$ , for each  $\ell$  (to arbitrary values). Such an event is a cylinder event on the digits.

**Theorem 6.9** (Filtration energy bound; proved). *In this situation let  $\lambda_\ell \geq 1$  ( $\ell \in \mathcal{P}$ ), and suppose that every event of  $\mathcal{E}$  is a digit-prefix event with  $\prod_\ell \lambda_\ell^{2v_\ell(E)} \leq 2$ . Then*

$$\sum_U \left( \prod_{\ell: U_\ell \neq \emptyset} \lambda_\ell^{1+\max U_\ell} \right) \|F^{=U}\|^2 \leq 1.$$

*Proof.* Write  $\Lambda_{\ell,i} = \lambda_\ell^{i+1}$ ,  $\Lambda_{\ell,i_0(\ell)-1} = 1$ , and  $\mu'_{\ell,j} = \Lambda_{\ell,j} - \Lambda_{\ell,j-1} \geq 0$  for  $i_0(\ell) \leq j < f_\ell$ . For such  $j$  let  $P_{\ell,j} = I - \mathbb{E}_{B_{\ell,j}}$ , where  $\mathbb{E}_{B_{\ell,j}}$  averages over the *tail block*  $B_{\ell,j} = (\delta_{\ell,i})_{i \geq j}$ . Averaging over a digit in  $U$  kills  $F^{=U}$ , and  $F^{=U}$  does not depend on digits outside  $U$ ; hence  $P_{\ell,j} F^{=U} = F^{=U}$  if  $U_\ell \neq \emptyset$  and  $\max U_\ell \geq j$ , and  $P_{\ell,j} F^{=U} = 0$  otherwise. Call a set  $V$  of pairs  $(\ell, j)$  a *selection* if it contains at most one pair for each  $\ell$ , and put  $L'_V = \prod_{(\ell,j) \in V} P_{\ell,j}$ . Then  $\|L'_V F\|^2 = \sum_{U \succ V} \|F^{=U}\|^2$ , where  $U \succ V$  means  $\max U_\ell \geq j$  for all  $(\ell, j) \in V$ . For fixed  $U$ , the sum of  $\mu'^V$  over the selections  $V \prec U$  factorises as  $\prod_\ell (1 + \sum_{i_0(\ell) \leq j \leq \max U_\ell} \mu'_{\ell,j}) = \prod_{\ell: U_\ell \neq \emptyset} \Lambda_{\ell, \max U_\ell}$ . So the left side of the claim equals  $\sum_V \mu'^V \|L'_V F\|^2$ , the sum over selections.

Fix a selection  $V$  and pass to the coarser product space whose coordinates are, for  $(\ell, j) \in V$ , the head  $(\delta_{\ell,i})_{i < j}$  and the tail block  $B_{\ell,j}$ , and, for the other  $\ell$ , the whole  $X_\ell$ . There  $L'_V$  is the operator  $L_{V'}$  of the Setting for the set  $V'$  of tail blocks. Each event  $E$  is a disjoint union of cylinder events on the coarse coordinates with the same coarse support (enumerate the unfixed digits of each coarse coordinate that  $E$  involves), and splitting every event in this way does not change  $F$ . Exactly one piece of  $E$  holds at  $x$  if  $E$  does. The trace of the coarse support of  $E$  on  $V'$  is the set of  $B_{\ell,j}$  with  $(\ell, j) \in V$  and  $j < v_\ell(E)$ . Identify  $B_{\ell,j}$  with the pair  $(\ell, j)$  and put  $\hat{E} = \{(\ell, j) : i_0(\ell) \leq j < v_\ell(E)\}$ . By Lemma 6.2 and the remark on traces,

$$\|L'_V F\|^2 \leq \mathbb{E}_x [N_{\hat{\mathcal{H}}(x)}(V)^2], \quad \hat{\mathcal{H}}(x) = \{\hat{E} : E \in \mathcal{E}, x \in E\}.$$

Summing over selections and adding the nonnegative terms of all other sets  $V$  of pairs, the left side of the claim is at most  $\mathbb{E}_x [Q_{\mu'}(\hat{\mathcal{H}}(x))]$ . By Theorem 6.5(a) it remains to show  $\prod_{(\ell,j) \in \hat{E}} (1 + \mu'_{\ell,j}) \leq 2$  for every event. Fix  $\ell$  and write  $\lambda = \lambda_\ell$ ,  $v = v_\ell(E)$ ,  $i_0 = i_0(\ell)$ , and assume  $v > i_0$ . Here

$1 + \mu'_{\ell, i_0} = \lambda^{i_0+1}$  and  $1 + \mu'_{\ell, j} = 1 + \lambda^j(\lambda - 1)$  for  $j > i_0$ . The hypothesis gives  $\lambda^{2v} \leq 2$ , so  $\lambda^j \leq \lambda^v \leq \sqrt{2} < \lambda + 1$  for  $j < v$ , i.e.  $1 + \lambda^j(\lambda - 1) \leq \lambda^2$ . The factor of  $\ell$  is therefore at most  $\lambda^{i_0+1} \lambda^{2(v-1-i_0)} \leq \lambda^{2v}$ , and the product over  $\ell$  is at most  $\prod_{\ell} \lambda_{\ell}^{2v_{\ell}(E)} \leq 2$ .  $\square$

**Corollary 6.10** (Modulus-weighted tail; proved). *In Theorem 6.9 let the  $\ell \in \mathcal{P}$  be primes and  $\lambda_{\ell} = 2^{\log \ell / (2\mathcal{L})}$ , and suppose  $\prod_{\ell} \ell^{v_{\ell}(E)} \leq T$  for every event. For a set  $U$  of digits put  $m_U = \prod_{\ell: U_{\ell} \neq \emptyset} \ell^{1+\max U_{\ell}}$ . Then for every  $\tau \geq 0$*

$$\sum_{U: m_U > e^{\tau}} \|F^{=U}\|^2 \leq 2^{-\tau/(2\mathcal{L})}.$$

*Proof.*  $\prod_{\ell} \lambda_{\ell}^{2v_{\ell}(E)} = 2^{\log \prod_{\ell} \ell^{v_{\ell}(E)} / \mathcal{L}} \leq 2$ , and the weight of  $U$  in Theorem 6.9 is  $2^{\log m_U / (2\mathcal{L})}$ .  $\square$

The cost of a junta is thus measured by the modulus  $m_U$  a function of the digits in  $U$  needs, not by the number of coordinates. For the Erdős–Straus events (digits in base  $\ell$ ) this removes both the binary width and the support size from the junta modulus.

## 7. THE RESIDUAL SYSTEM AND THE SANDWICH MINORANT

Throughout this section  $Q$  is a graded quarantine as in Definition 2.3,  $E_1, \dots, E_m$  ( $m \leq T^2$ ) are the events of the surviving atoms, in some order,  $A_i = \mathbf{1}_{E_i}$ ,  $F_{<i} = \prod_{j < i} (1 - A_j)$ ,  $F = F_{<m+1}$ ,  $w_{\ell} = w_{\ell}(Q)$ ,  $S = S(Q)$  and  $\delta = \mathbb{E}F$ . We assume

$$\sum_{\ell \in \text{supp } E_i} w_{\ell} \leq \frac{1}{64} \quad (i \leq m), \quad w_{\ell} \leq \frac{1}{64} \quad (\ell \in \mathcal{P}), \quad (7)$$

which holds for the output of Lemma 2.5 with  $c = 1/64$ .

*Digits.* For  $\ell \in \mathcal{P}$  write  $x \in \Omega_{\ell}$  in base  $\ell$  as  $x = \sum_{i < f_{\ell}} \delta_{\ell, i} \ell^i$ . The digits  $\delta_{\ell, i}$  with  $i < a_{\ell}$  are those of 1, hence constant on  $\Omega_{\ell}$ . The others are independent and uniform:  $\delta_{\ell, 0}$  on  $\{1, \dots, \ell - 1\}$  if  $a_{\ell} = 0$ , and  $\delta_{\ell, i}$  on  $\{0, \dots, \ell - 1\}$  for  $\max(a_{\ell}, 1) \leq i < f_{\ell}$ ; indeed  $x \mapsto (\delta_{\ell, i})_{i \geq a_{\ell}}$  is a bijection from  $\Omega_{\ell}$  onto the product of these ranges. So the free digits are those with  $i \geq i_0(\ell) := a_{\ell}$ , and  $E_{M,D}$  is a digit-prefix event with  $v_{\ell}(E_{M,D}) = v_{\ell}(M)$  for  $\ell \in \text{supp}(M, D)$  and  $v_{\ell}(E_{M,D}) = 0$  otherwise. A function of the digits  $\delta_{\ell, i}$  with  $i \leq i_{\ell}$  depends only on  $X_{\ell}$  modulo  $\ell^{1+i_{\ell}}$ .

*Cells.* A *consistent unit cell* is a class  $b \bmod d$  with  $d$  composed of primes of  $\mathcal{P}$ ,  $d \mid \prod_{\ell \in \mathcal{P}} \ell^{f_{\ell}}$ ,  $\gcd(b, d) = 1$  and  $b \equiv 1 \pmod{\gcd(d, Q)}$ ; on  $\Omega$  it is the event  $\{X_{\ell} \equiv b \pmod{\ell^{v_{\ell}(d)}} \forall \ell \mid d\}$ , and on  $\mathbb{Z}$  the class itself. The events  $E_{M,D}$  are consistent unit cells (by survival). The intersection of two consistent unit cells is empty or a consistent unit cell whose modulus is the lcm of the two, both on  $\Omega$  and on  $\mathbb{Z}$ . A function on  $\Omega$  that depends only on  $(X_{\ell} \bmod \ell^{b_{\ell}})_{\ell}$  is a linear combination of consistent unit cells of modulus  $\prod_{\ell} \ell^{b_{\ell}}$  (the indicators of its level sets' points).

**Lemma 7.1** (Haar means; proved). *Let  $Q' = Q\ell'$  with  $\ell'$  a prime not dividing  $Q \prod_{\ell \in \mathcal{P}} \ell$  (or  $Q' = Q$ ), and let  $\varphi$  be a combination of consistent unit cells with lcm of moduli  $D$ . With  $H$  as in Theorem 5.1 for  $Q'$  and  $D$ ,  $\mathbb{E}_H \varphi = \mathbb{E} \varphi$ . The same holds for  $\varphi\psi$ , where  $\psi$  is a character modulo a squarefree  $f \mid D$  coprime to  $Q$ , viewed on  $\Omega$  through  $X_\ell \bmod \ell$  for  $\ell \mid f$ .*

*Proof.* Let  $N_0 = \text{lcm}(Q', D)$ . By the Chinese remainder theorem the uniform measure on  $H$  is the product of the uniform measures on  $H_p = \{x \in (\mathbb{Z}/p^{v_p(N_0)})^\times : x \equiv 1 \pmod{p^{v_p(Q')}}\}$ . For  $\ell \in \mathcal{P}$  with  $\ell \mid D$ ,  $v_\ell(Q') = a_\ell$  and  $v_\ell(N_0) \leq f_\ell$ , and reduction maps the uniform measure on  $\Omega_\ell$  to the uniform measure on  $H_\ell$  (all fibres have the same size). The other primes of  $N_0$  do not divide  $D$ , and  $\varphi, \psi$  do not depend on them.  $\square$

**Lemma 7.2** (Local lemma with conditioning; classical). *Let  $\mathcal{A}$  be a finite family of events in a product space, each depending on a finite set of coordinates; write  $A' \sim A$  if they share a coordinate. Suppose  $x_A \in [0, 1)$  satisfy  $\mathbb{P}(A) \leq x_A \prod_{A' \sim A, A' \neq A} (1 - x_{A'})$ . Then  $\mathbb{P}(\bigcap_A \bar{A}) \geq \prod_A (1 - x_A)$ . Moreover, for every event  $B$  depending on a set  $J$  of coordinates,  $\mathbb{P}(B \mid \bigcap_A \bar{A}) \leq \mathbb{P}(B) \prod_{A \in \Gamma(B)} (1 - x_A)^{-1}$ , where  $\Gamma(B)$  is the set of  $A \in \mathcal{A}$  depending on some coordinate in  $J$ .*

*Proof.* The first statement is [1, Lemma 5.1.1]. Its proof shows  $\mathbb{P}(A \mid \bigcap_{A' \in \mathcal{S}} \bar{A}') \leq x_A$  for every  $\mathcal{S} \subseteq \mathcal{A}$  with  $A \notin \mathcal{S}$ . For the second (cf. [14, Thm. 2.1]), note that  $B$  is independent of  $\bigcap_{A \notin \Gamma(B)} \bar{A}$ , so

$$\mathbb{P}\left(B \mid \bigcap_A \bar{A}\right) \leq \frac{\mathbb{P}(B) \mathbb{P}(\bigcap_{A \notin \Gamma(B)} \bar{A})}{\mathbb{P}(\bigcap_A \bar{A})} = \frac{\mathbb{P}(B)}{\mathbb{P}(\bigcap_{\Gamma(B)} \bar{A} \mid \bigcap_{A \notin \Gamma(B)} \bar{A})}.$$

The chain rule and the displayed claim bound the denominator below by  $\prod_{\Gamma(B)} (1 - x_A)$ .  $\square$

**Corollary 7.3** (Proved). *Assume (7). Then  $\delta = \mathbb{E}F \geq e^{-2.2S}$ . Moreover, for every subfamily  $\mathcal{A}$  of  $\{E_1, \dots, E_m\}$  and every event  $B$  depending on the coordinates  $X_\ell$ ,  $\ell \in J$ , where  $\sum_{\ell \in J} w_\ell \leq 1/64$ ,  $\mathbb{P}(B \mid \text{no } A \in \mathcal{A} \text{ occurs}) \leq 1.04 \mathbb{P}(B)$ .*

*Proof.* Apply Lemma 7.2 with the coordinates  $X_\ell$  and  $x_{E_i} = 2\mathbb{P}(E_i) \leq 2w_\ell \leq 1/32$  for any  $\ell \in \text{supp } E_i$ . For an event  $B$  on  $J$ ,  $\sum_{A \in \Gamma(B)} x_A \leq 2 \sum_{\ell \in J} w_\ell$ . With  $B = E_i$  this is at most  $1/32$ , so  $\prod_{A \sim E_i, A \neq E_i} (1 - x_A) \geq 31/32$ , and the hypothesis of Lemma 7.2 holds for every subfamily. Since  $1 - x \geq e^{-1.1x}$  for  $0 \leq x \leq 1/32$ , we get  $\delta \geq \prod_i (1 - 2\mathbb{P}(E_i)) \geq e^{-2.2S}$ , and  $\prod_{\Gamma(B)} (1 - x_A)^{-1} \leq e^{1.1/32} \leq 1.04$  when  $\sum_{\ell \in J} w_\ell \leq 1/64$ .  $\square$

For  $j \leq m$  let  $F^{(j)}$  be the function of the digits not fixed by  $E_j$  obtained from  $F_{<j}$  by giving the digits fixed by  $E_j$  their values on  $E_j$ . Then  $A_j F_{<j} = A_j F^{(j)}$ . At  $\ell$ , the digits not fixed by  $E_j$  are those with  $i \geq i_0^{(j)}(\ell) := \max(a_\ell, v_\ell(E_j))$ , and  $F^{(j)}$  is the avoidance indicator of the restrictions of

the events  $E_i$  ( $i < j$ ) compatible with  $E_j$ . Each restriction fixes the digits  $i_0^{(j)}(\ell) \leq i < v_\ell(E_i)$ , so it is a digit-prefix event for the filtration starting at  $i_0^{(j)}$ , with the same  $v_\ell(E_i)$  and  $\prod_\ell \ell^{v_\ell(E_i)} \leq M_i \leq T$ . (A restriction may fix nothing; then  $F^{(j)} \equiv 0$ .)

**Lemma 7.4** (Bazzi–Razborov sandwich [3, 29]; proved). *Let  $u_1, \dots, u_m$  be arbitrary real functions,  $v_i = \sum_{j < i} A_j u_j$ , and*

$$B = 1 - \sum_{i \leq m} A_i (1 - v_i)^2.$$

*Then  $B \leq F$  pointwise, and with  $e_j = F_{<j} - u_j$ ,*

$$F - B = \sum_i A_i \left( \sum_{j < i} A_j e_j \right)^2, \quad \mathbb{E}[F - B] \leq m^2 \sum_{j \leq m} \mathbb{E}[A_j e_j^2].$$

*If  $u_j$  is a function of the digits not fixed by  $E_j$ , then  $\mathbb{E}[A_j e_j^2] = \mathbb{P}(E_j) \mathbb{E}(F^{(j)} - u_j)^2$ .*

*Proof.* The identity  $1 - F = \sum_i A_i F_{<i}$  holds pointwise: the sum counts the first occurring event. Put  $f_{<i} = 1 - F_{<i} = \sum_{j < i} A_j F_{<j}$ . If  $F_{<i} = 1$ , then  $A_j = 0$  for all  $j < i$ , so  $v_i = 0$  and  $A_i(1 - v_i)^2 = A_i = A_i F_{<i}$ . If  $F_{<i} = 0$ , then  $f_{<i} = 1$ , and  $A_i(1 - v_i)^2 - A_i F_{<i} = A_i(f_{<i} - v_i)^2$ . In both cases  $A_i(1 - v_i)^2 - A_i F_{<i} = A_i(f_{<i} - v_i)^2$ , and summing over  $i$ ,

$$F - B = \sum_i A_i [(1 - v_i)^2 - F_{<i}] = \sum_i A_i (f_{<i} - v_i)^2 \geq 0,$$

with  $f_{<i} - v_i = \sum_{j < i} A_j (F_{<j} - u_j)$ . By Cauchy–Schwarz,  $(\sum_{j < i} A_j e_j)^2 \leq m \sum_{j < i} A_j e_j^2$ . Drop  $A_i \leq 1$  and sum over  $i$ . For the last claim,  $A_j e_j^2 = A_j (F^{(j)} - u_j)^2$ , and  $F^{(j)} - u_j$  is independent of  $A_j$ .  $\square$

This is Razborov’s simplification [29] of Bazzi’s argument [3]. The choice of  $u_j$  as an  $\ell^2$ -approximation of the “conditional survival function”  $F^{(j)}$  is the one used there. We include the short proof for completeness. The identity and the inequality  $B \leq F$  use only that each  $A_i$  takes values in  $\{0, 1\}$ , so they also hold for the functions on  $\mathbb{Z}$  obtained from cell expansions.

*Choice of  $u_j$ .* Fix  $\tau \geq 0$ . For  $j \leq m$  let  $\mathcal{D}_j$  be the family of sets  $U$  of digits not fixed by  $E_j$  with  $m_U \leq e^\tau$ , where  $m_U = \prod_{\ell: U_\ell \neq \emptyset} \ell^{1 + \max U_\ell}$ ; it is down-closed. Put

$$u_j = (F^{(j)})^{\mathcal{D}_j} = \sum_{U \in \mathcal{D}_j} (F^{(j)})^{=U}, \quad (8)$$

the Efron–Stein decomposition being taken over the digits not fixed by  $E_j$ .

**Lemma 7.5** (Approximation error; proved). *With  $u_j$  as in (8),  $\mathbb{E}[F - B] \leq m^2 S 2^{-\tau/(2\mathcal{L})}$ .*

*Proof.* By (5) and Corollary 6.10, applied to  $F^{(j)}$  with the filtration starting at  $i_0^{(j)}$  (the digits are indexed by their absolute position, so  $m_U$  is the modulus needed),  $\mathbb{E}(F^{(j)} - u_j)^2 \leq 2^{-\tau/(2\mathcal{L})}$ . Conclude with Lemma 7.4.  $\square$

**Lemma 7.6** (Cell structure of  $B$ ; proved). *With  $u_j$  as in (8),  $B$  is a finite real combination of consistent unit cells of modulus at most  $T^3 e^{2\tau}$ , composed of primes of  $\mathcal{P}$ . The identity  $B = 1 - \sum_i A_i(1 - v_i)^2$  also holds for the expanded cell combination evaluated at any integer  $n$ .*

*Proof.* By (5),  $u_j$  is a combination of the functions  $\mathbb{E}[F^{(j)} \mid \delta_W]$ ,  $W \in \mathcal{D}_j$ . Such a function depends only on  $X_\ell \bmod \ell^{1+\max W_\ell}$  for the  $\ell$  with  $W_\ell \neq \emptyset$ , so it is a combination of consistent unit cells of modulus  $m_W \leq e^\tau$ . Expanding,

$$B = 1 - \sum_i A_i + 2 \sum_i \sum_{j < i} A_i A_j u_j - \sum_i \sum_{j, j' < i} A_i A_j A_{j'} u_j u_{j'}.$$

Each term is a product of at most three events (moduli  $\leq T$ ) and at most two  $u$ 's, so every cell is an intersection of consistent unit cells of total modulus at most  $T^3 e^{2\tau}$ . The last claim holds because the intersection identities hold on  $\mathbb{Z}$ .  $\square$

**Lemma 7.7** ( $\ell^1$ -tightness; proved). *If  $B \leq F$  pointwise,  $F \geq 0$  and  $\mathbb{E}[F - B] \leq \eta \mathbb{E}B$ , then  $\mathbb{E}|B| \leq (1 + 2\eta)\mathbb{E}B$ .*

*Proof.*  $B^- := \max(-B, 0) \leq F - B$  since  $F \geq 0$ , and  $\mathbb{E}|B| = \mathbb{E}B + 2\mathbb{E}B^-$ .  $\square$

**Lemma 7.8** (Twist; proved). *Assume (7) and  $\mathbb{E}[F - B] \leq \delta/100$ , and write  $B$  as in Lemma 7.6, with  $\mu = \mathbb{E}B$ . Let  $\psi$  be a real primitive character of conductor  $f > 1$  with  $\gcd(f, Q) = 1$  and  $f \mid d_i$  for some cell modulus  $d_i$ . Then  $\mu \geq 0.99\delta$  and  $|\mathbb{E}[B\psi]| \leq \mu/4$ , where  $\psi$  is viewed on  $\Omega$  through  $X_\ell \bmod \ell$ ,  $\ell \mid f$ .*

*Proof.*  $\mu = \mathbb{E}F - \mathbb{E}[F - B] \geq 0.99\delta$ . As  $2 \nmid f$ ,  $f$  is squarefree. Its prime factors divide  $d_i$ , so they lie in  $\mathcal{P}$ , and  $a_\ell = 0$  for them as  $\gcd(f, Q) = 1$ . So  $\psi = \prod_{\ell \mid f} \psi_\ell(X_\ell \bmod \ell)$  with nontrivial characters  $\psi_\ell$  of the uniformly distributed digits  $\delta_{\ell,0}$ . Now

$$|\mathbb{E}[B\psi]| \leq \mathbb{E}|B - F| + |\mathbb{E}[F\psi]| = \mathbb{E}[F - B] + |\mathbb{E}[F\psi]|.$$

Fix  $\ell_0 \mid f$  and let  $F'$  be the avoidance indicator of the events  $E_i$  with  $\ell_0 \notin \text{supp } E_i$ . Given the other coordinates  $X_{-\ell_0}$ , we have  $F = F' \cdot \mathbf{1}[X_{\ell_0} \notin \text{Forb}]$ , where  $\text{Forb}$  is the union of the classes  $X_{\ell_0} \equiv -4D_i \pmod{\ell_0^{v_{\ell_0}(M_i)}}$  over the events  $E_i \ni \ell_0$  whose other conditions hold. Write  $E_i \setminus \ell_0$  for the event given by those other conditions, and  $\pi_i$  for the  $\Omega_{\ell_0}$ -probability of the class at  $\ell_0$ , so that  $\mathbb{P}(E_i) = \mathbb{P}(E_i \setminus \ell_0)\pi_i$ . Since  $\mathbb{E}_{X_{\ell_0}} \psi_{\ell_0} = 0$ ,

$$\begin{aligned} |\mathbb{E}[F\psi]| &= \left| \mathbb{E} \left[ F' \prod_{\ell \neq \ell_0} \psi_\ell \mathbb{E}_{X_{\ell_0}} [\psi_{\ell_0} \mathbf{1}[X_{\ell_0} \in \text{Forb}]] \right] \right| \\ &\leq \sum_{i: \ell_0 \in \text{supp } E_i} \mathbb{P}(F' = 1, E_i \setminus \ell_0) \pi_i. \end{aligned}$$

The coordinates of  $E_i \setminus \ell_0$  lie in  $\text{supp } E_i$ , so by (7) and Corollary 7.3 applied to the family defining  $F'$ ,  $\mathbb{P}(E_i \setminus \ell_0 \mid F' = 1) \leq 1.04 \mathbb{P}(E_i \setminus \ell_0)$ . Hence  $|\mathbb{E}[F\psi]| \leq 1.04 w_{\ell_0} \mathbb{E}F' \leq 0.0163 \mathbb{E}F'$ . Likewise  $\mathbb{E}F' - \mathbb{E}F = \mathbb{E}[F' \mathbf{1}[X_{\ell_0} \in \text{Forb}]] \leq 0.0163 \mathbb{E}F'$ , so  $\mathbb{E}F' \leq \delta/0.9837$ . Altogether  $|\mathbb{E}[B\psi]| \leq 0.01\delta + 0.0166\delta \leq 0.027\mu/0.99 < \mu/4$ .  $\square$

## 8. PROOF OF THE MAIN THEOREMS

**Theorem 8.1** (Assembly; proved modulo Theorem 4.2). *Let  $T$  be large and let  $Q$  be the output of Lemma 2.5 with  $c = 1/64$ . Put  $S_1 = C_1 S_0$  and*

$$\tau = 2\mathcal{L} \lceil \log_2 (100 T^4 (S_1 + 1) e^{3S_1}) \rceil.$$

*Then there is a prime  $p \equiv 1 \pmod{840}$  with  $p > T$ ,  $W(p) > T$  and  $\log p \leq C_6 \log Z$ , where  $C_6$  is absolute and*

$$\log Z \leq \log Q + 4\tau + 6\mathcal{L} + 1.$$

*Proof.* Take the residual system of  $Q$ . By Lemma 2.5, (7) holds,  $S \leq S_1$  and  $840 \mid Q$ ; and  $m \leq T^2$ . If  $m = 0$ , then  $F \equiv 1$ ; we put  $B = 1$  (one cell of modulus 1), so that  $\mu = A = 1$  and the twist condition is vacuous, and go directly to the transfer step below. Otherwise define  $B$  by Lemma 7.4 with  $u_j$  as in (8). By Lemma 7.5, the choice of  $\tau$ ,  $m \leq T^2$  and Corollary 7.3,

$$\mathbb{E}[F - B] \leq m^2 S 2^{-\tau/(2\mathcal{L})} \leq \frac{S e^{-3S_1}}{100(S_1 + 1)} \leq \frac{e^{-3S}}{100} \leq \frac{\delta}{100}.$$

Lemma 7.8 gives  $\mu = \mathbb{E}B \geq 0.99\delta > 0$ , and, as  $\mathbb{E}[F - B] \leq \mu/99$ , Lemma 7.7 gives  $\mathbb{E}|B| \leq 1.03\mu$ . By Lemma 7.6,  $B$  is a combination of consistent unit cells of moduli  $d_i \leq T^3 e^{2\tau}$ .

*Pointwise inequality on integers.* For every integer  $n$  the values  $A_i(n) \in \{0, 1\}$  satisfy the hypotheses of Lemma 7.4, and the cell expansion of  $B$  is valid on  $\mathbb{Z}$  (Lemma 7.6). So  $B(n) \leq F(n)$ , and by Lemma 2.4(i),  $B(n) \leq 1[W(n) > T]$  for every  $n \equiv 1 \pmod{Q}$ .

*Transfer.* Let  $R = \max(T, \max_i d_i)$  and choose a prime  $\ell_{\text{aux}} \in (R, 2R]$ . Apply Theorem 5.1 with  $Q' = Q\ell_{\text{aux}}$ , so  $8 \mid Q'$ . As  $\ell_{\text{aux}} > \max d_i$ ,  $\gcd(d_i, Q') = \gcd(d_i, Q)$ , so the cells are consistent for  $Q'$ . The inequality  $B \leq 1[W > T]$  holds on  $n \equiv 1 \pmod{Q'}$ . By Lemma 7.1,  $\mathbb{E}_H B = \mu$  and  $\mathbb{E}_H |B|/\mu \leq 1.03 \leq Z^{1/4}$ . A real primitive character of conductor  $f > 1$  coprime to  $Q'$  is coprime to  $Q$ , so Lemmas 7.8 and 7.1 give the twist condition. Here  $\log Z \leq \log Q + \log(2R) + \log \max d_i \leq \log Q + \log 2 + 2(3\mathcal{L} + 2\tau)$ . Theorem 5.1 with  $\log x = C_2(1 + \log 1.03) \log Z$  gives a prime  $p \equiv 1 \pmod{Q'}$  with  $W(p) > T$  and  $\log p \leq C_6 \log Z$ . Since  $\ell_{\text{aux}} \mid p - 1$ ,  $p > \ell_{\text{aux}} > T$ , and  $840 \mid Q$  gives  $p \equiv 1 \pmod{840}$ .  $\square$

*Proof of Theorem 1.1.* By Theorem 3.7,  $S_1 \ll \mathcal{L}^4$  and  $\Omega_0 \ll \mathcal{L}^4 \log \mathcal{L}$ . Hence  $\tau \ll \mathcal{L}(\mathcal{L} + S_1) \ll \mathcal{L}^5$ , and Lemma 2.5(c) with  $c = 1/64$  gives  $\log Q \leq 1.02\mathcal{L} + 7 + 64C_1\mathcal{L}\Omega_0 \ll \mathcal{L}^5 \log \mathcal{L}$ . So  $\log Z \ll \mathcal{L}^5 \log \mathcal{L}$ , and Theorem 8.1 gives a hard prime  $p > T$  with  $W(p) > T$  and  $\log p \leq C\mathcal{L}^5 \log \mathcal{L}$ . As  $\mathcal{L} < \log p$ , this gives  $\mathcal{L}^5 \geq \log p / (C \log \log p)$ , i.e.  $W(p) > T \geq \exp(c(\log p)^{1/5}(\log \log p)^{-1/5})$ . Applying this to a sequence  $T_1 < T_2 < \dots$  with  $T_{j+1}$  larger than the prime found for  $T_j$  gives infinitely many distinct primes.  $\square$

*Proof of Theorem 1.2.* Use the unconditional part of Theorem 3.7. Now  $\tau$ ,  $\log Q$  and  $\log Z$  are all at most  $\mathcal{L}^{O(1)}(S_0 + 1)$ , so

$$\log \log p \leq \log(S_0 + 1) + O(\log \mathcal{L}) \leq (\log 2 + o(1)) \mathcal{L} / \log \mathcal{L}.$$

Since  $x \mapsto x/\log x$  is increasing for  $x \geq e$ , we get  $\mathcal{L}/\log \mathcal{L} \geq Y := (1/\log 2 - o(1)) \log \log p$ , hence  $\mathcal{L} \geq Y \log \mathcal{L} \geq Y \log Y$ . That is,  $\log W(p) > \mathcal{L} \geq (1/\log 2 - o(1)) \log \log p \cdot \log \log \log p$ . Infinitely many primes arise as before.  $\square$

**Corollary 8.2** (Haar side; proved modulo Theorem 3.1). *The set of integers  $n$  with  $W(n) > T$ , which is periodic, has density  $\delta^*(T)$  with  $\log(1/\delta^*(T)) \ll \mathcal{L}^5 \log \mathcal{L}$ .*

*Proof.* Take  $Q$  as in Theorem 8.1. The integers  $n \equiv 1 \pmod{Q}$  with  $n \bmod \ell^{f_\ell} \in \Omega_\ell$  for all  $\ell \in \mathcal{P}$  and  $F(n) = 1$  have  $W(n) > T$  (Lemma 2.4(i)). Their density is  $Q^{-1} \prod_{\ell \in \mathcal{P}, a_\ell=0} (1 - 1/\ell) \cdot \delta$ , and  $\delta \geq e^{-2.2S_1}$  by Corollary 7.3. So  $\log(1/\delta^*(T)) \leq \log Q + \log \mathcal{L} + O(1) + 2.2S_1 \ll \mathcal{L}^5 \log \mathcal{L}$  by Mertens' theorem, Lemma 2.5 and Theorem 3.7. Gallagher's theorem is not used.  $\square$

**Remark 8.3** (Where the exponent comes from; Assessment). The certified bound is  $\log p \ll \log Z \ll \log Q + \tau$ . The junta term is  $\tau \asymp \mathcal{L}(S_1 + \mathcal{L}) \ll \mathcal{L}^5$ , and the quarantine term is  $\log Q \ll \mathcal{L} \Omega_0 \ll \mathcal{L}^5 \log \mathcal{L}$ . In the previous version these were  $\mathcal{L}^7$  (bit width  $\asymp \mathcal{L}$ , support size  $k$ , and  $k_0 \asymp S$  coordinates at cost  $\mathcal{L}$  each) and  $\mathcal{L}^7/\log \mathcal{L}$  (full quarantine of a block of small primes and of  $\ll k^2 S$  bad primes). The remaining factor  $\log \mathcal{L}$  comes from the moment  $\Omega_0$ ; we do not know whether  $\Omega_0 \ll \mathcal{L}^4$ . Both terms are of order  $\mathcal{L} \cdot S$  under natural heuristics: any graded class-of-one quarantine feeding the local-lemma criterion used here needs small fibre masses at every coordinate, and on model systems the junta modulus needed by an Efron–Stein truncation is of order  $\mathcal{L}S$ . So the exponent  $1/5$  appears to be the natural limit of the present route, but we have not proved lower bounds for either term. We make no claim about the true order of  $W(p)$ .

## 9. SUMMARY OF STATUS

- Proved: Lemmas 2.1, 2.2, 2.4, 2.5, 3.2, 3.3; the energy bound (Lemmas 6.2–6.4, Theorems 6.5 and 6.9, Corollaries 6.6 and 6.10); Lemmas 7.1, 7.4, 7.5, 7.6, 7.7, 7.8 and Corollary 7.3 (with the classical Lemma 7.2); the unconditional part of Theorem 3.7 (with the classical divisor bound).
- Proved modulo the Elsholtz–Tao estimates (Theorem 3.1): Lemmas 3.4, 3.5, 3.6, Theorem 3.7 and the Haar-side Corollary 8.2.
- Proved modulo Gallagher's theorem (Theorems 4.2 and 4.1, quoted from the draft [22]): the linear transfer Theorem 5.1, Theorem 8.1 and Theorem 1.2.
- Proved modulo Gallagher's theorem and Theorem 3.1: Theorem 1.1.
- Assessment: Remarks 5.2, 3.8 and 8.3; the numerical parts of Remarks 6.7 and 6.8.
- Not claimed: anything about the Erdős–Straus conjecture; optimality of the exponent  $1/5$ ; any numerical instance; priority for Theorem 6.5 (literature search partial).

- Håstad’s switching lemma and the Thorner–Zaman theorem are no longer used; they are cited only for comparison.

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